

```
In [ ]: !pip install roboflow

from roboflow import Roboflow
rf = Roboflow(api_key="8s40mZrnk8wBMxLkmHT1")
project = rf.workspace("neural-networks-research").project("hard-hat-detection-6
version = project.version(1)
dataset = version.download("yolov11")
```

Collecting roboflow

Downloading roboflow-1.3.8-py3-none-any.whl.metadata (11 kB)

Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from roboflow) (2026.4.22)

Collecting idna==3.7 (from roboflow)

Downloading idna-3.7-py3-none-any.whl.metadata (9.9 kB)

Requirement already satisfied: cyclor in /usr/local/lib/python3.12/dist-packages (from roboflow) (0.12.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from roboflow) (1.5.0)

Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from roboflow) (3.10.0)

Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from roboflow) (2.0.2)

Collecting opencv-python-headless==4.10.0.84 (from roboflow)

Downloading opencv_python_headless-4.10.0.84-cp37-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (20 kB)

Requirement already satisfied: Pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from roboflow) (11.3.0)

Collecting pi-heif<2 (from roboflow)

Downloading pi_heif-1.3.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.3 kB)

Collecting pillow-avif-plugin<2 (from roboflow)

Downloading pillow_avif_plugin-1.5.5-cp312-cp312-manylinux_2_28_x86_64.whl.metadata (2.2 kB)

Requirement already satisfied: python-dateutil in /usr/local/lib/python3.12/dist-packages (from roboflow) (2.9.0.post0)

Requirement already satisfied: python-dotenv in /usr/local/lib/python3.12/dist-packages (from roboflow) (1.2.2)

Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from roboflow) (2.32.4)

Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages (from roboflow) (1.17.0)

Requirement already satisfied: urllib3>=1.26.6 in /usr/local/lib/python3.12/dist-packages (from roboflow) (2.5.0)

Requirement already satisfied: tqdm>=4.41.0 in /usr/local/lib/python3.12/dist-packages (from roboflow) (4.67.3)

Requirement already satisfied: PyYAML>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from roboflow) (6.0.3)

Requirement already satisfied: requests_toolbelt in /usr/local/lib/python3.12/dist-packages (from roboflow) (1.0.0)

Collecting filetype (from roboflow)

Downloading filetype-1.2.0-py2.py3-none-any.whl.metadata (6.5 kB)

Requirement already satisfied: typer>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from roboflow) (0.24.2)

Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer>=0.12.0->roboflow) (8.3.3)

Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.12.0->roboflow) (1.5.4)

Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.12.0->roboflow) (13.9.4)

Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.12.0->roboflow) (0.0.4)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->roboflow) (1.3.3)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->roboflow) (4.62.1)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->roboflow) (26.1)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist

```

-packages (from matplotlib->roboflow) (3.3.2)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python
3.12/dist-packages (from requests->roboflow) (3.4.7)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.1
2/dist-packages (from rich>=12.3.0->typer>=0.12.0->roboflow) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.
12/dist-packages (from rich>=12.3.0->typer>=0.12.0->roboflow) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packa
ges (from markdown-it-py>=2.2.0->rich>=12.3.0->typer>=0.12.0->roboflow) (0.1.2)
Downloading roboflow-1.3.8-py3-none-any.whl (207 kB)
_____ 207.9/207.9 kB 9.1 MB/s eta 0:00:00
Downloading idna-3.7-py3-none-any.whl (66 kB)
_____ 66.8/66.8 kB 3.3 MB/s eta 0:00:00
Downloading opencv_python_headless-4.10.0.84-cp37-abi3-manylinux_2_17_x86_64.many
linux2014_x86_64.whl (49.9 MB)
_____ 49.9/49.9 MB 15.9 MB/s eta 0:00:00
Downloading pi_heif-1.3.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_6
4.whl (1.5 MB)
_____ 1.5/1.5 MB 61.5 MB/s eta 0:00:00
Downloading pillow_avif_plugin-1.5.5-cp312-cp312-manylinux_2_28_x86_64.whl (5.5 M
B)
_____ 5.5/5.5 MB 58.9 MB/s eta 0:00:00
Downloading filetype-1.2.0-py2.py3-none-any.whl (19 kB)
Installing collected packages: pillow-avif-plugin, filetype, pi-heif, opencv-pyth
on-headless, idna, roboflow
  Attempting uninstall: opencv-python-headless
    Found existing installation: opencv-python-headless 4.13.0.92
    Uninstalling opencv-python-headless-4.13.0.92:
      Successfully uninstalled opencv-python-headless-4.13.0.92
  Attempting uninstall: idna
    Found existing installation: idna 3.13
    Uninstalling idna-3.13:
      Successfully uninstalled idna-3.13
Successfully installed filetype-1.2.0 idna-3.7 opencv-python-headless-4.10.0.84 p
i-heif-1.3.0 pillow-avif-plugin-1.5.5 roboflow-1.3.8
loading Roboflow workspace...
loading Roboflow project...
Downloading Dataset Version Zip in Hard-Hat-Detection-1 to yolov11:: 100%|
| 287964/287964 [00:05<00:00, 55211.55it/s]
Extracting Dataset Version Zip to Hard-Hat-Detection-1 in yolov11:: 100%|
| 10005/10005 [00:01<00:00, 5449.15it/s]

```

```

In [ ]: !pip install ultralytics
        from ultralytics import YOLO

        !ls

```

Collecting ultralytics

Downloading ultralytics-8.4.49-py3-none-any.whl.metadata (39 kB)

Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.2)

Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (3.10.0)

Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (4.13.0.92)

Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (11.3.0)

Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (6.0.3)

Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.32.4)

Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.16.3)

Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.10.0+cu128)

Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (0.25.0+cu128)

Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (5.9.5)

Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.35.2)

Collecting ultralytics-thop>=2.0.18 (from ultralytics)

Downloading ultralytics_thop-2.0.19-py3-none-any.whl.metadata (14 kB)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (4.62.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.5.0)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (26.1)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (3.3.2)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (2.9.0.post0)

Requirement already satisfied: polars-runtime-32==1.35.2 in /usr/local/lib/python3.12/dist-packages (from polars>=0.20.0->ultralytics) (1.35.2)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.4.7)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.7)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (2026.4.22)

Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.29.0)

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (4.15.0)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (75.2.0)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.6.1)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.1.6)

Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (2025.3.0)

Requirement already satisfied: cuda-bindings==12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.9.4)

Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.8.93)

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.8.90)

Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.8.90)

Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (9.10.2.21)

Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.8.4.1)

Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (11.3.3.83)

Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (10.3.9.90)

Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (11.7.3.90)

Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (0.7.1)

Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (2.27.5)

Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.4.5)

Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.8.90)

Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.8.93)

Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (1.13.1.3)

Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.6.0)

Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings==12.9.4->torch>=1.8.0->ultralytics) (1.5.3)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.3.0->ultralytics) (1.17.0)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.8.0->ultralytics) (1.3.0)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.8.0->ultralytics) (3.0.3)

Downloading ultralytics-8.4.49-py3-none-any.whl (1.2 MB)

1.2/1.2 MB 29.5 MB/s eta 0:00:00

Downloading ultralytics_thop-2.0.19-py3-none-any.whl (28 kB)

Installing collected packages: ultralytics-thop, ultralytics

Successfully installed ultralytics-8.4.49 ultralytics-thop-2.0.19

Creating new Ultralytics Settings v0.0.6 file 

View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'

Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see <https://docs.ultralytics.com/quickstart/#ultralytics-settings>.

Hard-Hat-Detection-1 sample_data

Cek Jumlah Perkelas

```
In [ ]: import os

train_path = "/content/Hard-Hat-Detection-1/train/images"
valid_path = "/content/Hard-Hat-Detection-1/valid/images"
test_path = "/content/Hard-Hat-Detection-1/test/images"

print("Train:", len(os.listdir(train_path)))
print("Valid:", len(os.listdir(valid_path)))
print("Test:", len(os.listdir(test_path)))
```

Train: 3500

Valid: 1000

Test: 500

Total keseluruhan data adalah 5000 dan dilakukan split sebesar 70:20:10 dengan jumlah data training sebanyak 3500, validation data sebanyak 1000, dan test sebanyak 500. Hasil print jumlah split sesuai dengan split otomatis yang sudah dilakukan oleh roboflow lalu bisa dilanjutkan ke proses selanjutnya.

Size Check

```
In [ ]: from PIL import Image

image_path = "/content/Hard-Hat-Detection-1/train/images"

files = os.listdir(image_path)[:5]

for file in files:
    img = Image.open(os.path.join(image_path, file))
    print(file, ":", img.size)
```

```
hard_hat_workers896_png.rf.9d20de144749939a45c9b19bacc03bfd.jpg : (640, 640)
hard_hat_workers1647_png.rf.617401e85e0e56943fd9f918db99bfbb.jpg : (640, 640)
hard_hat_workers987_png.rf.fdbf9e7f1598888dc99716f99729536f.jpg : (640, 640)
hard_hat_workers1615_png.rf.bbda9dad7a9a45c9aeda8ec239ddc4e7.jpg : (640, 640)
hard_hat_workers271_png.rf.f9c815fc99e5293d6135b9045868ad7e.jpg : (640, 640)
```

```
In [ ]: from PIL import Image

image_path = "/content/Hard-Hat-Detection-1/valid/images"

files = os.listdir(image_path)[:5]

for file in files:
    img = Image.open(os.path.join(image_path, file))
    print(file, ":", img.size)
```

```
hard_hat_workers1834_png.rf.78e6a1fed9013e2c3eca7a1f6edd2f32.jpg : (640, 640)
hard_hat_workers4015_png.rf.b998c16302f4d25acc468d1447fbc821.jpg : (640, 640)
hard_hat_workers222_png.rf.4f780bbe5617637939a0fcd31c3db245.jpg : (640, 640)
hard_hat_workers1060_png.rf.41658483ccaa965f1dfa01728daf7d3f.jpg : (640, 640)
hard_hat_workers809_png.rf.35319dec6941ca22ddc98bc4a9a76c9f.jpg : (640, 640)
```

```
In [ ]: from PIL import Image

image_path = "/content/Hard-Hat-Detection-1/test/images"

files = os.listdir(image_path)[:5]

for file in files:
    img = Image.open(os.path.join(image_path, file))
    print(file, ":", img.size)
```

```
hard_hat_workers3607_png.rf.8f9ba58ed9cec2396a38eebc370077cf.jpg : (640, 640)
hard_hat_workers2217_png.rf.9f8c64fd42324742b9c17124c1a3e7b9.jpg : (640, 640)
hard_hat_workers1457_png.rf.92c3f5d8502f61e8b360bd3d6b683843.jpg : (640, 640)
hard_hat_workers2178_png.rf.6358c03c5e2fe4fa51ad9430d60fc2c9.jpg : (640, 640)
hard_hat_workers816_png.rf.05af62e46e2414527b11abeb88f3a500.jpg : (640, 640)
```

Semua images yang ada di tiap dataset ini sudah diatur oleh roboflow agar memiliki ukuran images yang sama sehingga tidak diperlukan proses resizing image lagi. Kami pun melanjutkan ke proses selanjutnya.

Class Checking

```
In [ ]: with open('/content/Hard-Hat-Detection-1/data.yaml', 'r') as f:
        print(f.read())
```

```
train: ../train/images
val: ../valid/images
test: ../test/images
```

```
nc: 3
names: ['head', 'helmet', 'person']
```

```
roboflow:
  workspace: neural-networks-research
  project: hard-hat-detection-62dxm
  version: 1
  license: CC BY 4.0
  url: https://universe.roboflow.com/neural-networks-research/hard-hat-detection-62dxm/dataset/1
```

Model memiliki 3 kelas yang akan didetect, yaitu : head, helmet, dan person.

- Head adalah kepala orang (tanpa melihat pake helmet/tidak)
- Helmet adalah hard hat
- Person adalah keseluruhan tubuh

Visualisasi Sampel Dataset

```
In [ ]: import random
import matplotlib.pyplot as plt

image_dir = "/content/Hard-Hat-Detection-1/train/images"

images = random.sample(os.listdir(image_dir), 4)
```

```
plt.figure(figsize=(12,5))

for i, img_name in enumerate(images):
    img_path = os.path.join(image_dir, img_name)
    img = Image.open(img_path)

    plt.subplot(1,4,i+1)
    plt.imshow(img)
    plt.title(img_name[:10])
    plt.axis('off')

plt.show()
```



```
In [ ]: import random
import matplotlib.pyplot as plt

image_dir = "/content/Hard-Hat-Detection-1/valid/images"

images = random.sample(os.listdir(image_dir), 4)

plt.figure(figsize=(12,5))

for i, img_name in enumerate(images):
    img_path = os.path.join(image_dir, img_name)
    img = Image.open(img_path)

    plt.subplot(1,4,i+1)
    plt.imshow(img)
    plt.title(img_name[:10])
    plt.axis('off')

plt.show()
```



```
In [ ]: import random
import matplotlib.pyplot as plt

image_dir = "/content/Hard-Hat-Detection-1/test/images"

images = random.sample(os.listdir(image_dir), 4)
```



```
plt.figure(figsize=(12,5))

for i, img_name in enumerate(images):
    img_path = os.path.join(image_dir, img_name)
    img = Image.open(img_path)

    plt.subplot(1,4,i+1)
    plt.imshow(img)
    plt.title(img_name[:10])
    plt.axis('off')

plt.show()
```



Kode - Kode ini digunakan untuk melihat tiap images yang ada di train, test, dan valid yang digenerate secara random, hanya untuk melihat secara sekilas. Secara sekilas dapat dilihat bahwa ukuran tiap images sudah sama semua.

Distribusi Perkelas

```
In [ ]: import matplotlib.pyplot as plt
import yaml
import os

data_yaml_path = '/content/Hard-Hat-Detection-1/data.yaml'
with open(data_yaml_path, 'r') as f:
    data_config = yaml.safe_load(f)

classes = data_config['names']
num_classes = data_config['nc']

class_counts = {i: 0 for i in range(num_classes)}

base_data_dir = '/content/Hard-Hat-Detection-1'
label_dirs = [
    os.path.join(base_data_dir, 'train', 'labels'),
    os.path.join(base_data_dir, 'valid', 'labels'),
    os.path.join(base_data_dir, 'test', 'labels')
]

for label_dir in label_dirs:
    if not os.path.exists(label_dir):
        print(f"Warning: Label directory not found: {label_dir}. Skipping.")
        continue
    for label_file_name in os.listdir(label_dir):
        if label_file_name.endswith('.txt'):
            label_file_path = os.path.join(label_dir, label_file_name)
```

```

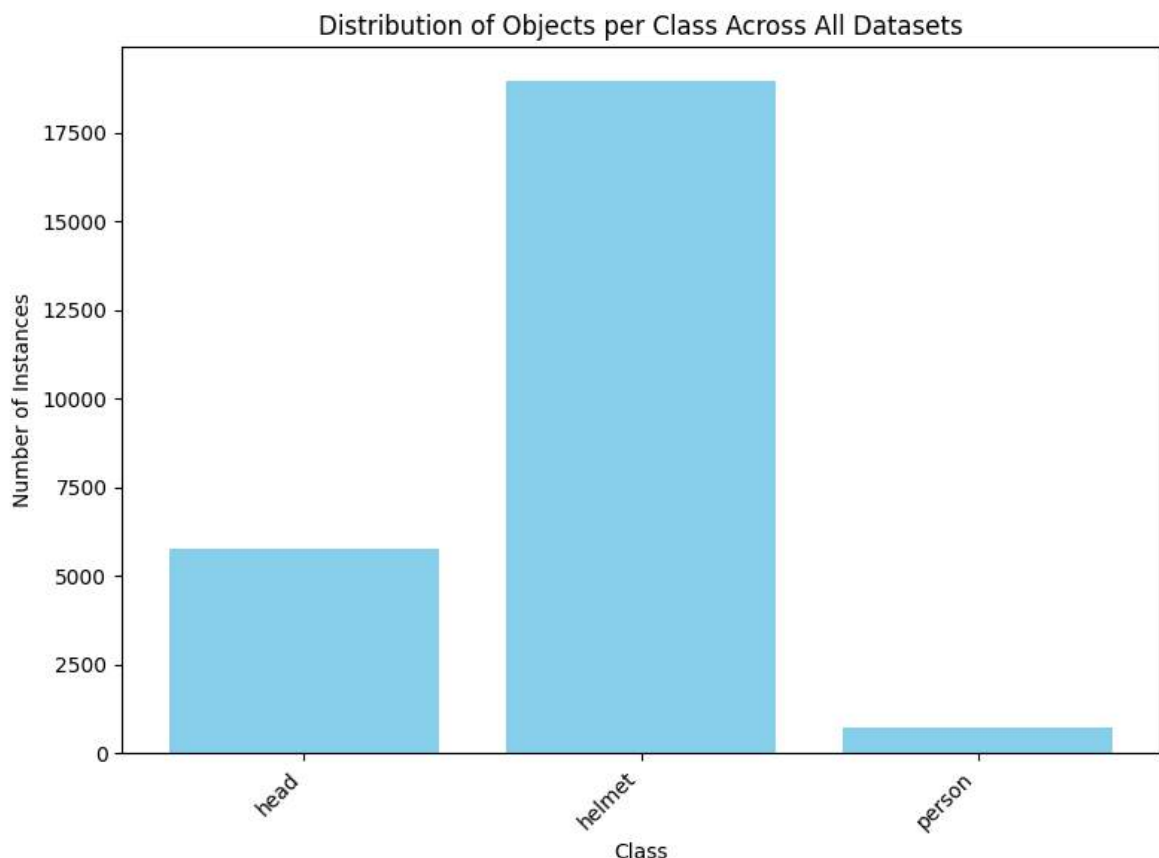
with open(label_file_path, 'r') as f:
    for line in f:
        parts = line.strip().split()
        if parts:
            class_id = int(parts[0])
            if 0 <= class_id < num_classes:
                class_counts[class_id] += 1
            else:
                print(f"Warning: Invalid class ID {class_id} found i

class_names = [classes[i] for i in sorted(class_counts.keys())]
counts = [class_counts[i] for i in sorted(class_counts.keys())]

# Plot the distribution
plt.figure(figsize=(8, 6))
plt.bar(class_names, counts, color='skyblue')
plt.xlabel("Class")
plt.ylabel("Number of Instances")
plt.title("Distribution of Objects per Class Across All Datasets")
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

print("Class distribution:")
for i, name in enumerate(class_names):
    print(f"- {name}: {counts[i]} instances")

```



Class distribution:

- head: 5785 instances
- helmet: 18966 instances
- person: 751 instances

Dari hasil grafik distribusi class secara keseluruhan, terdapat imbalance di dataset ini.

Dimana class helmet berjumlah sangat banyak yaitu 18966, sedangkan Head ada

sebanyak 5785, dan person hanya sebanyak 751.

```
In [ ]: import matplotlib.pyplot as plt
import yaml
import os

data_yaml_path = '/content/Hard-Hat-Detection-1/data.yaml'
with open(data_yaml_path, 'r') as f:
    data_config = yaml.safe_load(f)

classes = data_config['names']
num_classes = data_config['nc']

class_counts = {i: 0 for i in range(num_classes)}

base_data_dir = '/content/Hard-Hat-Detection-1'

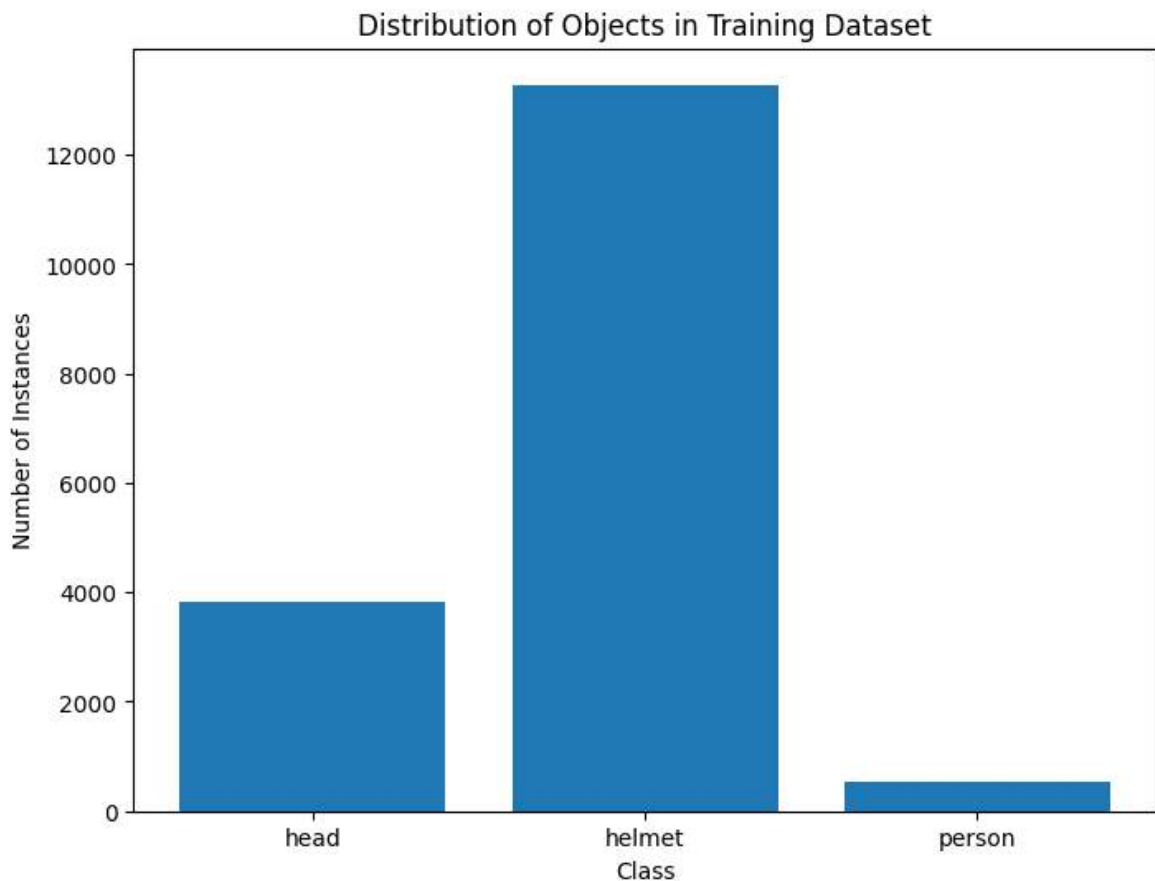
# TRAIN ONLY
label_dirs = [
    os.path.join(base_data_dir, 'train', 'labels')
]

for label_dir in label_dirs:
    for label_file_name in os.listdir(label_dir):
        if label_file_name.endswith('.txt'):
            label_file_path = os.path.join(label_dir, label_file_name)
            with open(label_file_path, 'r') as f:
                for line in f:
                    parts = line.strip().split()
                    if parts:
                        class_id = int(parts[0])
                        class_counts[class_id] += 1

class_names = [classes[i] for i in sorted(class_counts.keys())]
counts = [class_counts[i] for i in sorted(class_counts.keys())]

plt.figure(figsize=(8,6))
plt.bar(class_names, counts)
plt.xlabel("Class")
plt.ylabel("Number of Instances")
plt.title("Distribution of Objects in Training Dataset")
plt.show()

print("Training set distribution:")
for i, name in enumerate(class_names):
    print(f"{name}: {counts[i]}")
```



Training set distribution:

head: 3818

helmet: 13283

person: 540

Berdasarkan distribusi objek pada training dataset, juga terdapat imbalance jumlah anotasi antar kelas. Kelas helmet memiliki 13.283 , head memiliki 3.818, sedangkan person memiliki jumlah paling sedikit. Imbalance ini akan menyebabkan model lebih banyak belajar mengenali kelas mayoritas sehingga performa deteksi pada kelas person menjadi rendah.

Images Size and Resolution

```
In [ ]: sizes = []

image_data_dirs = [
    '/content/Hard-Hat-Detection-1/train/images',
    '/content/Hard-Hat-Detection-1/valid/images',
    '/content/Hard-Hat-Detection-1/test/images'
]

for img_dir in image_data_dirs:
    if not os.path.exists(img_dir):
        print(f"Warning: Image directory not found: {img_dir}. Skipping.")
        continue
    # Take a sample, e.g., first 50 images from each directory
    for img_name in os.listdir(img_dir)[:50]: # Limiting to 50 for quick check
        img_path = os.path.join(img_dir, img_name)
        try:
            img = Image.open(img_path)
```

```

        sizes.append(img.size)
    except Exception as e:
        print(f"Error opening image {img_name} in {img_dir}: {e}")

# Check for consistency if sizes were collected
if len(sizes) > 0:
    first_size = sizes[0]
    all_sample_same_size = all(s == first_size for s in sizes)
    print(f"All sampled images have the same size ({first_size}): {all_sample_sa
else:
    print("No images processed to check sizes.")

print("Sample of image sizes (first 10 collected):")
for s in sizes[:10]:
    print(s)

```

All sampled images have the same size ((640, 640)): True

Sample of image sizes (first 10 collected):

```

(640, 640)
(640, 640)
(640, 640)
(640, 640)
(640, 640)
(640, 640)
(640, 640)
(640, 640)
(640, 640)
(640, 640)

```

In []: `import pandas as pd`

```

df_sizes = pd.DataFrame(sizes, columns=['width', 'height'])
print(df_sizes.describe())

```

	width	height
count	150.0	150.0
mean	640.0	640.0
std	0.0	0.0
min	640.0	640.0
25%	640.0	640.0
50%	640.0	640.0
75%	640.0	640.0
max	640.0	640.0

semua gambar sudah memiliki ukuran yang sama yaitu 640x640

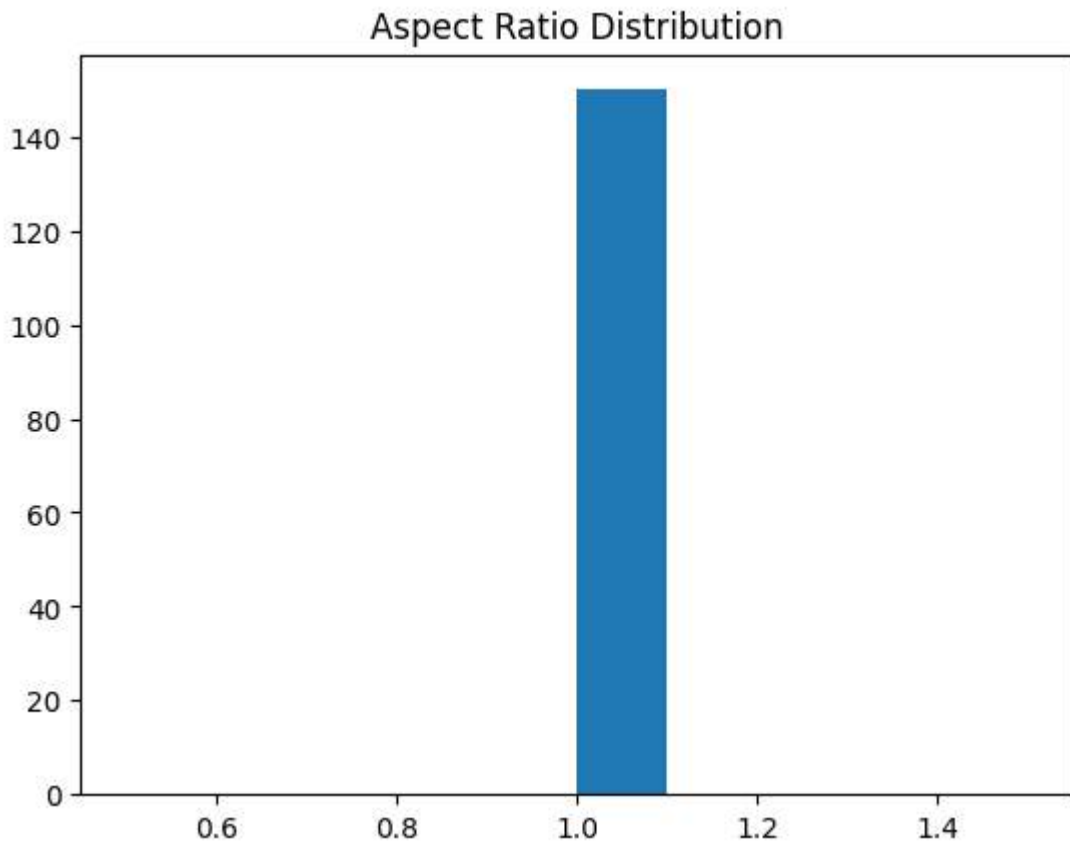
Ratio Distribution

```

In [ ]: ratios = [w/h for w,h in sizes]

plt.hist(ratios)
plt.title("Aspect Ratio Distribution")
plt.show()

```



Semua images memiliki rasio yang sudah seragam, yaitu 1.

CHECK : image corrupt, jumlah object per-image, channel rgb,ratio

```
In [ ]: import os
import cv2
import numpy as np
import pandas as pd
from PIL import Image
from tqdm import tqdm

def advanced_edu(dataset_path, split='train'):
    image_folder = os.path.join(dataset_path, split, 'images')
    label_folder = os.path.join(dataset_path, split, 'labels')

    data_stats = []
    corrupt_files = []

    for img_name in tqdm(os.listdir(image_folder)):
        img_path = os.path.join(image_folder, img_name)
        lab_path = os.path.join(label_folder, img_name.replace('.jpg', '.txt').r

        # Cek Image Corrupt & Channel
        try:
            with Image.open(img_path) as img:
                width, height = img.size
                channels = len(img.getbands()) # RGB = 3, Grayscale = 1, RGBA =
```

```

# Cek 3 channel (RGB)
is_rgb = (channels == 3)

# Cek Jumlah Objek
obj_count = 0
if os.path.exists(lab_path):
    with open(lab_path, 'r') as f:
        obj_count = len(f.readlines())

data_stats.append({
    'file': img_name,
    'width': width,
    'height': height,
    'aspect_ratio': round(width/height, 2),
    'channels': channels,
    'obj_count': obj_count
})
except Exception as e:
    corrupt_files.append(img_name)

df = pd.DataFrame(data_stats)

print("\n--- HASIL EDA ---")
print(f"File Corrupt: {len(corrupt_files)}")
print(f"Rata-rata Objek per Gambar: {df['obj_count'].mean():.2f}")
print(f"Gambar Tanpa Objek (Background): {len(df[df['obj_count'] == 0])}")
print(f"Min Dimension: {df['width'].min()}x{df['height'].min()}")
print(f"Max Dimension: {df['width'].max()}x{df['height'].max()}")
print(f"Konsistensi RGB (Gambar 3-Channel): {len(df[df['channels'] == 3])} dari {len(df)}")

extreme_ar = df[(df['aspect_ratio'] > 2) | (df['aspect_ratio'] < 0.5)]
print(f"Gambar dengan Aspect Ratio Ekstrem: {len(extreme_ar)}")

return df

df_eda = advanced_eda(dataset.location, 'train')

```

```
100%|██████████| 3500/3500 [00:00<00:00, 4505.23it/s]
```

```
--- HASIL EDA ---
```

```
File Corrupt: 0
```

```
Rata-rata Objek per Gambar: 5.04
```

```
Gambar Tanpa Objek (Background): 0
```

```
Min Dimension: 640x640
```

```
Max Dimension: 640x640
```

```
Konsistensi RGB (Gambar 3-Channel): 3500 dari 3500 gambar
```

```
Gambar dengan Aspect Ratio Ekstrem: 0
```

Dari hasil diatas, tidak ditemukan file yang corrupt, sehingga seluruh gambar dapat langsung dimanfaatkan tanpa perlu ada yang dihapus maupun diperbaiki terlebih dahulu. Ukuran gambar pun seragam di resolusi 640×640, sehingga model dapat menerima input yang konsisten selama training. Setiap gambar rata-rata mengandung sekitar 5 objek, sehingga model mendapat kesempatan untuk belajar mendeteksi lebih dari satu objek dalam satu frame secara bersamaan. Di samping itu, tidak ditemukan gambar yang hanya berisi latar belakang kosong, artinya seluruh data memang mengandung objek-objek yang relevan dan bernilai untuk proses training. Dari sisi format, semua gambar telah menggunakan RGB 3-channel. Hal ini dapat menjaga konsistensi dataset secara keseluruhan serta meminimalkan risiko distorsi saat gambar

diproses oleh model. Secara umum, dataset layak digunakan langsung untuk tahap pelatihan model.

Visualisasi Data

check label dan kualitas image

```
In [ ]: import cv2
import matplotlib.pyplot as plt
import random
import os

def show_sample(split='train'):
    image_dir = os.path.join(dataset.location, split, 'images')
    label_dir = os.path.join(dataset.location, split, 'labels')

    img_name = random.choice(os.listdir(image_dir))
    img_path = os.path.join(image_dir, img_name)
    label_path = os.path.join(label_dir, img_name.replace('.jpg', '.txt').replace(' ', '_'))

    image = cv2.imread(img_path)
    image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    h, w, _ = image.shape

    if os.path.exists(label_path):
        with open(label_path, 'r') as f:
            for line in f:
                cls, x, y, bw, bh = map(float, line.split())

                x1 = int((x - bw/2) * w)
                y1 = int((y - bh/2) * h)
                x2 = int((x + bw/2) * w)
                y2 = int((y + bh/2) * h)

                cv2.rectangle(image, (x1, y1), (x2, y2), (0, 255, 0), 2)
                cv2.putText(image, class_names[int(cls)], (x1, y1-10),
                            cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 255, 0), 2)

    plt.figure(figsize=(10, 10))
    plt.imshow(image)
    plt.title(f"Visualisasi data: {img_name}")
    plt.axis('off')
    plt.show()

show_sample('train')
```


Visualisasi data: hard_hat_workers1542_png.rf.96d7a53133484001b5f5370b2bbedf40.jpg



Dari hasil visualisasi, bounding box berhasil mengelilingi objek seperti head/helmet dan label pada bounding box sudah sesuai. Ini berarti label sudah terbaca dengan benar dan format anotasi YOLO sudah sesuai.

Training Menggunakan Model YOLO

Untuk project dataset object detection ini, kelompok kami menggunakan model YOLOv11n dengan framework Ultralytics. YOLOv11n cocok digunakan karena model ini dirancang untuk inferensi cepat sehingga ideal untuk kebutuhan monitoring real-time. Selain itu, objek hard hat tergolong sederhana secara visual dengan bentuk yang khas, sehingga arsitektur nano yang lebih ringan sudah cukup untuk mempelajari polanya tanpa memerlukan model yang besar dan kompleks. Input native YOLOv11n yang 640×640 juga sesuai dengan ukuran gambar pada dataset, sehingga tidak diperlukan resize ulang yang dapat menurunkan kualitas. Dengan rata-rata 5 objek per gambar, kemampuan multi-object detection YOLO dalam satu forward pass pun dapat dimanfaatkan secara optimal.

```
In [ ]: model = YOLO("yolo11n.pt")

model.train(
    data="/content/Hard-Hat-Detection-1/data.yaml",
    epochs=50,
    imgsz=640,
    patience=10
)
```

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo11n.pt> to 'yolo11n.pt': 100% ————— 5.4MB 68.8MB/s 0.1s

Ultralytics 8.4.49 🚀 Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)

engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugment, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close_mosaic=10, cls=0.5, cls_pw=0.0, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=/content/Hard-Hat-Detection-1/data.yaml, degrees=0.0, deterministic=True, device=None, dfl=1.5, dnn=False, dropout=0.0, dynamic=False, embed=None, end2end=None, epochs=50, erasing=0.4, exist_ok=False, flip_lr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None, half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras=False, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolo11n.pt, momentum=0.937, mosaic=1.0, multi_scale=0.0, name=train, nbs=64, nms=False, opset=None, optimize=False, optimizer=auto, overlap_mask=True, patience=10, perspective=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=None, rect=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False, save_crop=False, save_dir=/content/runs/detect/train, save_frames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True, single_cls=False, source=None, split=val, stream_buffer=False, task=detect, time=None, tracker=botsort.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None

Downloading <https://ultralytics.com/assets/Arial.ttf> to '/root/.config/Ultralytics/Arial.ttf': 100% ————— 755.1KB 21.8MB/s 0.0s

Overriding model.yaml nc=80 with nc=3

	from	n	params	module
arguments				
0	-1	1	464	ultralytics.nn.modules.conv.Conv
[3, 16, 3, 2]				
1	-1	1	4672	ultralytics.nn.modules.conv.Conv
[16, 32, 3, 2]				
2	-1	1	6640	ultralytics.nn.modules.block.C3k2
[32, 64, 1, False, 0.25]				
3	-1	1	36992	ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]				
4	-1	1	26080	ultralytics.nn.modules.block.C3k2
[64, 128, 1, False, 0.25]				
5	-1	1	147712	ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]				
6	-1	1	87040	ultralytics.nn.modules.block.C3k2
[128, 128, 1, True]				
7	-1	1	295424	ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]				
8	-1	1	346112	ultralytics.nn.modules.block.C3k2
[256, 256, 1, True]				
9	-1	1	164608	ultralytics.nn.modules.block.SPPF
[256, 256, 5]				
10	-1	1	249728	ultralytics.nn.modules.block.C2PSA
[256, 256, 1]				
11	-1	1	0	torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']				
12	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat
[1]				
13	-1	1	111296	ultralytics.nn.modules.block.C3k2
[384, 128, 1, False]				
14	-1	1	0	torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']				

```

15          [-1, 4] 1          0 ultralytics.nn.modules.conv.Concat
[1]
16          -1 1      32096 ultralytics.nn.modules.block.C3k2
[256, 64, 1, False]
17          -1 1      36992 ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]
18          [-1, 13] 1          0 ultralytics.nn.modules.conv.Concat
[1]
19          -1 1      86720 ultralytics.nn.modules.block.C3k2
[192, 128, 1, False]
20          -1 1     147712 ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
21          [-1, 10] 1          0 ultralytics.nn.modules.conv.Concat
[1]
22          -1 1     378880 ultralytics.nn.modules.block.C3k2
[384, 256, 1, True]
23          [16, 19, 22] 1     431257 ultralytics.nn.modules.head.Detect
[3, 16, None, [64, 128, 256]]
YOLO11n summary: 182 layers, 2,590,425 parameters, 2,590,409 gradients, 6.4 GFLOPs

```

Transferred 448/499 items from pretrained weights

Freezing layer 'model.23.dfl.conv.weight'

AMP: running Automatic Mixed Precision (AMP) checks...

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo26n.pt> to 'yolo26n.pt': 100% ————— 5.3MB 85.0MB/s 0.1s

AMP: checks passed 

train: Fast image access  (ping: 0.0±0.0 ms, read: 1757.5±1102.3 MB/s, size: 60.8 KB)

train: Scanning /content/Hard-Hat-Detection-1/train/labels... 3500 images, 0 backgrounds, 0 corrupt: 100% ————— 3500/3500 2.1Kit/s 1.6s

train: New cache created: /content/Hard-Hat-Detection-1/train/labels.cache

augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average', num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

val: Fast image access  (ping: 0.0±0.0 ms, read: 523.7±279.9 MB/s, size: 57.0 KB)

val: Scanning /content/Hard-Hat-Detection-1/valid/labels... 1000 images, 0 backgrounds, 0 corrupt: 100% ————— 1000/1000 797.1it/s 1.3s

val: New cache created: /content/Hard-Hat-Detection-1/valid/labels.cache

optimizer: 'optimizer=auto' found, ignoring 'lr=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr' and 'momentum' automatically...

optimizer: AdamW(lr=0.001429, momentum=0.9) with parameter groups 81 weight(decay=0.0), 88 weight(decay=0.0005), 87 bias(decay=0.0)

Plotting labels to /content/runs/detect/train/labels.jpg...

Image sizes 640 train, 640 val

Using 2 dataloader workers

Logging results to /content/runs/detect/train

Starting training for 50 epochs...

	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
	1/50	2.57G	1.66	2.002	1.26	116	640: 10
0%	—————	219/219	2.4it/s	1:32			
		Class	Images	Instances	Box(P	R	mAP50 mA
P50-95):	100%	—————	32/32	1.9it/s	16.7s		
		all	1000	5175	0.879	0.46	0.516
							0.282

	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
	2/50	2.73G	1.574	1.324	1.232	84	640: 10

0%	219/219	3.4it/s	1:04				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.7it/s	8.6s				
	all	1000	5175	0.886	0.456	0.515	
0.274							
Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size	
3/50	2.73G	1.589	1.224	1.242	103	640: 10	
0%	219/219	3.4it/s	1:05				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.6it/s	8.9s				
	all	1000	5175	0.818	0.389	0.462	
0.239							
Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size	
4/50	2.73G	1.559	1.142	1.231	71	640: 10	
0%	219/219	3.5it/s	1:03				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.5it/s	9.1s				
	all	1000	5175	0.874	0.485	0.54	
0.29							
Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size	
5/50	2.73G	1.538	1.074	1.229	57	640: 10	
0%	219/219	3.4it/s	1:05				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.5it/s	9.1s				
	all	1000	5175	0.901	0.511	0.566	
0.307							
Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size	
6/50	2.73G	1.511	1.019	1.205	119	640: 10	
0%	219/219	3.3it/s	1:05				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.7it/s	8.6s				
	all	1000	5175	0.91	0.494	0.553	
0.311							
Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size	
7/50	2.73G	1.505	1.007	1.203	86	640: 10	
0%	219/219	3.3it/s	1:06				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.5it/s	9.1s				
	all	1000	5175	0.9	0.501	0.558	
0.329							
Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size	
8/50	2.73G	1.497	0.9895	1.202	71	640: 10	
0%	219/219	3.4it/s	1:04				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	2.4it/s	13.5s				
	all	1000	5175	0.926	0.517	0.574	
0.322							
Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size	
9/50	2.73G	1.501	0.9659	1.197	131	640: 10	
0%	219/219	3.0it/s	1:14				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.6it/s	8.8s				
	all	1000	5175	0.912	0.539	0.593	

0.335

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
10/50	2.73G	1.46	0.9382	1.186	56	640: 10
0% —————	219/219	3.4it/s	1:05			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	—————	32/32	4.0it/s	8.0s		
	all	1000	5175	0.922	0.521	0.579

0.329

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
11/50	2.73G	1.461	0.9264	1.181	66	640: 10
0% —————	219/219	3.4it/s	1:05			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	—————	32/32	3.5it/s	9.1s		
	all	1000	5175	0.93	0.535	0.594

0.34

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
12/50	2.73G	1.458	0.9168	1.179	80	640: 10
0% —————	219/219	3.4it/s	1:04			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	—————	32/32	3.9it/s	8.2s		
	all	1000	5175	0.915	0.547	0.594

0.35

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
13/50	2.73G	1.437	0.8993	1.165	120	640: 10
0% —————	219/219	3.4it/s	1:04			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	—————	32/32	3.5it/s	9.2s		
	all	1000	5175	0.924	0.541	0.595

0.355

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
14/50	2.73G	1.437	0.8833	1.165	63	640: 10
0% —————	219/219	3.4it/s	1:04			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	—————	32/32	3.9it/s	8.2s		
	all	1000	5175	0.927	0.544	0.6

0.357

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
15/50	2.73G	1.429	0.8787	1.158	91	640: 10
0% —————	219/219	3.4it/s	1:04			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	—————	32/32	3.5it/s	9.2s		
	all	1000	5175	0.901	0.536	0.582

0.337

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
16/50	2.91G	1.42	0.8558	1.154	65	640: 10
0% —————	219/219	3.5it/s	1:03			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	—————	32/32	3.8it/s	8.3s		
	all	1000	5175	0.928	0.558	0.608

0.359

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
17/50	2.91G	1.426	0.8636	1.162	88	640: 10

0%	219/219	3.2it/s	1:09				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.5it/s	9.1s				
	all	1000	5175	0.918	0.557	0.602	
0.347							
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	18/50	2.91G	1.415	0.8586	1.151	93	640: 10
0%	219/219	3.1it/s	1:10				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.5it/s	9.2s				
	all	1000	5175	0.927	0.55	0.606	
0.36							
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	19/50	2.91G	1.401	0.8449	1.15	73	640: 10
0%	219/219	3.4it/s	1:04				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.4it/s	9.5s				
	all	1000	5175	0.928	0.563	0.609	
0.37							
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	20/50	2.91G	1.388	0.8233	1.144	69	640: 10
0%	219/219	3.5it/s	1:03				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.5it/s	9.1s				
	all	1000	5175	0.931	0.563	0.614	
0.362							
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	21/50	2.91G	1.395	0.8237	1.143	73	640: 10
0%	219/219	3.4it/s	1:05				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.7it/s	8.6s				
	all	1000	5175	0.933	0.565	0.614	
0.368							
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	22/50	3.1G	1.388	0.8265	1.14	77	640: 10
0%	219/219	3.5it/s	1:02				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.4it/s	9.4s				
	all	1000	5175	0.933	0.553	0.612	
0.363							
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	23/50	3.1G	1.38	0.8212	1.136	87	640: 10
0%	219/219	3.4it/s	1:04				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.8it/s	8.3s				
	all	1000	5175	0.936	0.564	0.616	
0.365							
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	24/50	3.1G	1.381	0.8114	1.136	144	640: 10
0%	219/219	3.4it/s	1:05				
	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	3.4it/s	9.3s				
	all	1000	5175	0.912	0.559	0.601	

0.361

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
25/50	3.1G	1.372	0.8056	1.134	107	640: 10
0% —————	219/219	3.5it/s	1:03			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32	4.0it/s	8.1s			
	all	1000	5175	0.933	0.562	0.616

0.369

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
26/50	3.1G	1.371	0.798	1.132	96	640: 10
0% —————	219/219	3.5it/s	1:03			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32	3.6it/s	9.0s			
	all	1000	5175	0.936	0.558	0.614

0.358

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
27/50	3.1G	1.365	0.7885	1.127	101	640: 10
0% —————	219/219	3.5it/s	1:03			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32	4.0it/s	8.1s			
	all	1000	5175	0.927	0.555	0.609

0.352

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
28/50	3.1G	1.361	0.7922	1.127	81	640: 10
0% —————	219/219	3.4it/s	1:04			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32	3.6it/s	8.8s			
	all	1000	5175	0.939	0.569	0.615

0.366

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
29/50	3.1G	1.347	0.7776	1.118	79	640: 10
0% —————	219/219	3.2it/s	1:08			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32	3.0it/s	10.5s			
	all	1000	5175	0.934	0.569	0.618

0.376

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
30/50	3.1G	1.354	0.7799	1.122	104	640: 10
0% —————	219/219	3.5it/s	1:03			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32	3.5it/s	9.2s			
	all	1000	5175	0.929	0.572	0.618

0.364

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
31/50	3.1G	1.336	0.7665	1.114	129	640: 10
0% —————	219/219	3.4it/s	1:04			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32	3.6it/s	8.9s			
	all	1000	5175	0.929	0.573	0.625

0.377

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
32/50	3.1G	1.358	0.776	1.116	52	640: 10

0%	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	33/50	3.1G	1.33	0.7632	1.115	67	640: 10
0%	219/219	3.4it/s	1:04				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	32/32	3.4it/s	9.5s				
	all	1000	5175	0.938	0.564	0.615	
0.365							
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	34/50	3.1G	1.33	0.7587	1.114	38	640: 10
0%	219/219	3.4it/s	1:04				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	32/32	3.9it/s	8.2s				
	all	1000	5175	0.939	0.566	0.618	
0.368							
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	35/50	3.1G	1.33	0.7408	1.107	66	640: 10
0%	219/219	3.4it/s	1:04				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	32/32	4.0it/s	8.0s				
	all	1000	5175	0.941	0.567	0.617	
0.371							
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	36/50	3.1G	1.32	0.7476	1.104	66	640: 10
0%	219/219	3.4it/s	1:04				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	32/32	3.3it/s	9.6s				
	all	1000	5175	0.932	0.574	0.616	
0.379							
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	37/50	3.1G	1.324	0.7421	1.102	68	640: 10
0%	219/219	3.4it/s	1:05				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	32/32	3.9it/s	8.3s				
	all	1000	5175	0.942	0.571	0.624	
0.379							
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	38/50	3.1G	1.305	0.7243	1.099	98	640: 10
0%	219/219	3.4it/s	1:04				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	32/32	3.5it/s	9.1s				
	all	1000	5175	0.937	0.573	0.617	
0.375							
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	39/50	3.1G	1.305	0.7382	1.103	74	640: 10
0%	219/219	3.4it/s	1:04				
	Class	Images	Instances	Box(P)	R	mAP50	mA
P50-95): 100%	32/32	4.1it/s	7.8s				
	all	1000	5175	0.932	0.577	0.619	

0.376

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	40/50	3.1G	1.297	0.7263	1.093	94	640: 10
0%	219/219 3.4it/s 1:05						
	Class		Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32 3.7it/s 8.7s						
	all		1000	5175	0.942	0.569	0.626

0.381

Closing dataloader mosaic

alumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average', num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	41/50	3.1G	1.291	0.6548	1.11	56	640: 10
0%	219/219 3.4it/s 1:05						
	Class		Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32 4.0it/s 8.0s						
	all		1000	5175	0.934	0.577	0.623

0.379

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	42/50	3.1G	1.277	0.6385	1.102	54	640: 10
0%	219/219 3.0it/s 1:13						
	Class		Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32 3.1it/s 10.5s						
	all		1000	5175	0.928	0.572	0.617

0.369

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	43/50	3.1G	1.273	0.6346	1.099	51	640: 10
0%	219/219 3.0it/s 1:12						
	Class		Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32 3.2it/s 10.0s						
	all		1000	5175	0.93	0.579	0.62

0.383

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	44/50	3.1G	1.273	0.634	1.096	50	640: 10
0%	219/219 3.1it/s 1:11						
	Class		Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32 3.1it/s 10.2s						
	all		1000	5175	0.937	0.573	0.618

0.367

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	45/50	3.1G	1.253	0.626	1.092	42	640: 10
0%	219/219 3.3it/s 1:07						
	Class		Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32 4.2it/s 7.7s						
	all		1000	5175	0.928	0.58	0.615

0.378

	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
	46/50	3.1G	1.261	0.621	1.09	69	640: 10
0%	219/219 3.4it/s 1:05						
	Class		Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32 2.6it/s 12.2s						
	all		1000	5175	0.938	0.573	0.621

0.378

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
47/50	3.1G	1.247	0.6138	1.092	39	640: 10
0% —————	219/219	2.7it/s	1:20			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32	2.9it/s	10.9s			
	all	1000	5175	0.932	0.581	0.621

0.384

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
48/50	3.1G	1.242	0.6147	1.087	52	640: 10
0% —————	219/219	3.3it/s	1:07			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32	3.2it/s	10.0s			
	all	1000	5175	0.938	0.577	0.621

0.376

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
49/50	3.1G	1.23	0.6049	1.082	77	640: 10
0% —————	219/219	3.5it/s	1:02			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32	3.8it/s	8.5s			
	all	1000	5175	0.939	0.575	0.624

0.383

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
50/50	3.1G	1.229	0.6062	1.079	42	640: 10
0% —————	219/219	3.5it/s	1:03			
	Class	Images	Instances	Box(P	R	mAP50 mA
P50-95): 100%	32/32	3.8it/s	8.4s			
	all	1000	5175	0.936	0.576	0.622

0.379

50 epochs completed in 1.054 hours.

Optimizer stripped from /content/runs/detect/train/weights/last.pt, 5.5MB

Optimizer stripped from /content/runs/detect/train/weights/best.pt, 5.5MB

Validating /content/runs/detect/train/weights/best.pt...

Ultralytics 8.4.49 🚀 Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)

YOLO11n summary (fused): 101 layers, 2,582,737 parameters, 0 gradients, 6.3 GFLOPs

	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%	32/32	2.9it/s	11.1s				
	all	1000	5175	0.932	0.581	0.621	
0.384	head	185	1203	0.876	0.839	0.882	
0.535	helmet	929	3826	0.92	0.903	0.95	
0.599	person	40	146	1	0	0.0329	

0.0169

Speed: 0.2ms preprocess, 2.3ms inference, 0.0ms loss, 2.5ms postprocess per image

Results saved to /content/runs/detect/train

Out[]: ultralytics.utils.metrics.DetMetrics object with attributes:

```

ap_class_index: array([0, 1, 2])
box: ultralytics.utils.metrics.Metric object
confusion_matrix: <ultralytics.utils.metrics.ConfusionMatrix object at 0x78bec8095c40>
curves: ['Precision-Recall(B)', 'F1-Confidence(B)', 'Precision-Confidence(B)', 'Recall-Confidence(B)']
curves_results: [[array([
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6927,	0.27027,	0.27127,	0.27227,	0.27327,	0.27427,	0.27
528,	0.27628,	0.27728,	0.27828,	0.27928,	0.28028,	0.281
28,	0.28228,	0.28328,	0.28428,	0.28529,	0.28629,	0.2872
9,						
	0.28829,	0.28929,	0.29029,	0.29129,	0.29229,	0.2
9329,	0.29429,	0.2953,	0.2963,	0.2973,	0.2983,	0.2
993,	0.3003,	0.3013,	0.3023,	0.3033,	0.3043,	0.305
31,	0.30631,	0.30731,	0.30831,	0.30931,	0.31031,	0.3113
1,						
	0.31231,	0.31331,	0.31431,	0.31532,	0.31632,	0.3
1732,	0.31832,	0.31932,	0.32032,	0.32132,	0.32232,	0.32
332,	0.32432,	0.32533,	0.32633,	0.32733,	0.32833,	0.329
33,	0.33033,	0.33133,	0.33233,	0.33333,	0.33433,	0.3353
4,						
	0.33634,	0.33734,	0.33834,	0.33934,	0.34034,	0.3
4134,	0.34234,	0.34334,	0.34434,	0.34535,	0.34635,	0.34
735,	0.34835,	0.34935,	0.35035,	0.35135,	0.35235,	0.353
35,	0.35435,	0.35536,	0.35636,	0.35736,	0.35836,	0.3593
6,						
	0.36036,	0.36136,	0.36236,	0.36336,	0.36436,	0.3
6537,	0.36637,	0.36737,	0.36837,	0.36937,	0.37037,	0.37
137,	0.37237,	0.37337,	0.37437,	0.37538,	0.37638,	0.377
38,	0.37838,	0.37938,	0.38038,	0.38138,	0.38238,	0.3833
8,						
	0.38438,	0.38539,	0.38639,	0.38739,	0.38839,	0.3
8939,	0.39039,	0.39139,	0.39239,	0.39339,	0.39439,	0.3
954,	0.3964,	0.3974,	0.3984,	0.3994,	0.4004,	0.40
14,	0.4024,	0.4034,	0.4044,	0.40541,	0.40641,	0.4074
1,						
	0.40841,	0.40941,	0.41041,	0.41141,	0.41241,	0.4
1341,	0.41441,	0.41542,	0.41642,	0.41742,	0.41842,	0.41
942,	0.42042,	0.42142,	0.42242,	0.42342,	0.42442,	0.425
43,	0.42643,	0.42743,	0.42843,	0.42943,	0.43043,	0.4314
3,						
	0.43243,	0.43343,	0.43443,	0.43544,	0.43644,	0.4
3744,	0.43844,	0.43944,	0.44044,	0.44144,	0.44244,	0.44
344,	0.44444,	0.44545,	0.44645,	0.44745,	0.44845,	0.449
45,	0.45045,	0.45145,	0.45245,	0.45345,	0.45445,	0.4554
6,						
	0.45646,	0.45746,	0.45846,	0.45946,	0.46046,	0.4
6146,	0.46246,	0.46346,	0.46446,	0.46547,	0.46647,	0.46
747,	0.46847,	0.46947,	0.47047,	0.47147,	0.47247,	0.473
47,	0.47447,	0.47548,	0.47648,	0.47748,	0.47848,	0.4794
8,						
	0.48048,	0.48148,	0.48248,	0.48348,	0.48448,	0.4
8549,	0.48649,	0.48749,	0.48849,	0.48949,	0.49049,	0.49
149,	0.49249,	0.49349,	0.49449,	0.4955,	0.4965,	0.49
75,	0.4985,	0.4995,	0.5005,	0.5015,	0.5025,	0.503
5,						
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0951,	0.51051,	0.51151,	0.51251,	0.51351,	0.51451,	0.51
552,	0.51652,	0.51752,	0.51852,	0.51952,	0.52052,	0.521
52,	0.52252,	0.52352,	0.52452,	0.52553,	0.52653,	0.5275
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197,      0.97297,      0.97397,      0.97497,      0.97598,      0.97698,      0.977
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0.024024,      0.025025,      0.026026,      0.027027,      0.028028,      0.02
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035,      0.036036,      0.037037,      0.038038,      0.039039,      0.04004,      0.0410
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059,      0.06006,      0.061061,      0.062062,      0.063063,      0.064064,      0.0650
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7077,      0.078078,      0.079079,      0.08008,      0.081081,      0.082082,      0.083
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7317,	0.17417,	0.17518,	0.17618,	0.17718,	0.17818,	0.17
918,	0.18018,	0.18118,	0.18218,	0.18318,	0.18418,	0.185
19,	0.18619,	0.18719,	0.18819,	0.18919,	0.19019,	0.1911
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032,	0.2042,	0.20521,	0.20621,	0.20721,	0.20821,	0.209
21,	0.21021,	0.21121,	0.21221,	0.21321,	0.21421,	0.2152
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	0.21622,	0.21722,	0.21822,	0.21922,	0.22022,	0.2
2122,	0.22222,	0.22322,	0.22422,	0.22523,	0.22623,	0.22
723,	0.22823,	0.22923,	0.23023,	0.23123,	0.23223,	0.233
23,	0.23423,	0.23524,	0.23624,	0.23724,	0.23824,	0.2392
4,						
	0.24024,	0.24124,	0.24224,	0.24324,	0.24424,	0.2
4525,	0.24625,	0.24725,	0.24825,	0.24925,	0.25025,	0.25
125,	0.25225,	0.25325,	0.25425,	0.25526,	0.25626,	0.257
26,	0.25826,	0.25926,	0.26026,	0.26126,	0.26226,	0.2632
6,						
	0.26426,	0.26527,	0.26627,	0.26727,	0.26827,	0.2
6927,	0.27027,	0.27127,	0.27227,	0.27327,	0.27427,	0.27
528,	0.27628,	0.27728,	0.27828,	0.27928,	0.28028,	0.281
28,	0.28228,	0.28328,	0.28428,	0.28529,	0.28629,	0.2872
9,						
	0.28829,	0.28929,	0.29029,	0.29129,	0.29229,	0.2
9329,	0.29429,	0.2953,	0.2963,	0.2973,	0.2983,	0.2
993,	0.3003,	0.3013,	0.3023,	0.3033,	0.3043,	0.305
31,	0.30631,	0.30731,	0.30831,	0.30931,	0.31031,	0.3113
1,						
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1732,	0.31832,	0.31932,	0.32032,	0.32132,	0.32232,	0.32
332,	0.32432,	0.32533,	0.32633,	0.32733,	0.32833,	0.329
33,	0.33033,	0.33133,	0.33233,	0.33333,	0.33433,	0.3353
4,						
	0.33634,	0.33734,	0.33834,	0.33934,	0.34034,	0.3
4134,	0.34234,	0.34334,	0.34434,	0.34535,	0.34635,	0.34
735,	0.34835,	0.34935,	0.35035,	0.35135,	0.35235,	0.353
35,	0.35435,	0.35536,	0.35636,	0.35736,	0.35836,	0.3593
6,						
	0.36036,	0.36136,	0.36236,	0.36336,	0.36436,	0.3
6537,	0.36637,	0.36737,	0.36837,	0.36937,	0.37037,	0.37
137,	0.37237,	0.37337,	0.37437,	0.37538,	0.37638,	0.377
38,	0.37838,	0.37938,	0.38038,	0.38138,	0.38238,	0.3833
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954,	0.3964,	0.3974,	0.3984,	0.3994,	0.4004,	0.40
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942,	0.42042,	0.42142,	0.42242,	0.42342,	0.42442,	0.425
43,	0.42643,	0.42743,	0.42843,	0.42943,	0.43043,	0.4314
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3744,	0.43844,	0.43944,	0.44044,	0.44144,	0.44244,	0.44
344,	0.44444,	0.44545,	0.44645,	0.44745,	0.44845,	0.449
45,	0.45045,	0.45145,	0.45245,	0.45345,	0.45445,	0.4554
6,						
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6146,	0.46246,	0.46346,	0.46446,	0.46547,	0.46647,	0.46
747,	0.46847,	0.46947,	0.47047,	0.47147,	0.47247,	0.473
47,	0.47447,	0.47548,	0.47648,	0.47748,	0.47848,	0.4794
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149,	0.49249,	0.49349,	0.49449,	0.4955,	0.4965,	0.49
75,	0.4985,	0.4995,	0.5005,	0.5015,	0.5025,	0.503
5,						
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0951,	0.51051,	0.51151,	0.51251,	0.51351,	0.51451,	0.51
552,	0.51652,	0.51752,	0.51852,	0.51952,	0.52052,	0.521
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55,	0.54655,	0.54755,	0.54855,	0.54955,	0.55055,	0.5515
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356,	0.56456,	0.56557,	0.56657,	0.56757,	0.56857,	0.569
57,	0.57057,	0.57157,	0.57257,	0.57357,	0.57457,	0.5755
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8158,	0.58258,	0.58358,	0.58458,	0.58559,	0.58659,	0.58
759,	0.58859,	0.58959,	0.59059,	0.59159,	0.59259,	0.593
59,	0.59459,	0.5956,	0.5966,	0.5976,	0.5986,	0.599
6,						
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0561,	0.60661,	0.60761,	0.60861,	0.60961,	0.61061,	0.61
161,	0.61261,	0.61361,	0.61461,	0.61562,	0.61662,	0.617
62,	0.61862,	0.61962,	0.62062,	0.62162,	0.62262,	0.6236
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2963,	0.63063,	0.63163,	0.63263,	0.63363,	0.63463,	0.63
564,	0.63664,	0.63764,	0.63864,	0.63964,	0.64064,	0.641
64,	0.64264,	0.64364,	0.64464,	0.64565,	0.64665,	0.6476
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966,	0.66066,	0.66166,	0.66266,	0.66366,	0.66466,	0.665
67,	0.66667,	0.66767,	0.66867,	0.66967,	0.67067,	0.6716
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368,	0.68468,	0.68569,	0.68669,	0.68769,	0.68869,	0.689
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7,						
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771,	0.70871,	0.70971,	0.71071,	0.71171,	0.71271,	0.713
71,	0.71471,	0.71572,	0.71672,	0.71772,	0.71872,	0.7197
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74,	0.73874,	0.73974,	0.74074,	0.74174,	0.74274,	0.7437
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576,	0.75676,	0.75776,	0.75876,	0.75976,	0.76076,	0.761
76,	0.76276,	0.76376,	0.76476,	0.76577,	0.76677,	0.7677
7,						
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7377,	0.77477,	0.77578,	0.77678,	0.77778,	0.77878,	0.77
978,	0.78078,	0.78178,	0.78278,	0.78378,	0.78478,	0.785
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2963,	0.63063,	0.63163,	0.63263,	0.63363,	0.63463,	0.63
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978,	0.78078,	0.78178,	0.78278,	0.78378,	0.78478,	0.785
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	0.79279,	0.79379,	0.79479,	0.7958,	0.7968,	0.
7978,	0.7988,	0.7998,	0.8008,	0.8018,	0.8028,	0.8
038,	0.8048,	0.80581,	0.80681,	0.80781,	0.80881,	0.809

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81,      0.81081,      0.81181,      0.81281,      0.81381,      0.81481,      0.8158
2,
      0.81682,      0.81782,      0.81882,      0.81982,      0.82082,      0.8
2182,      0.82282,      0.82382,      0.82482,      0.82583,      0.82683,      0.82
783,      0.82883,      0.82983,      0.83083,      0.83183,      0.83283,      0.833
83,      0.83483,      0.83584,      0.83684,      0.83784,      0.83884,      0.8398
4,
      0.84084,      0.84184,      0.84284,      0.84384,      0.84484,      0.8
4585,      0.84685,      0.84785,      0.84885,      0.84985,      0.85085,      0.85
185,      0.85285,      0.85385,      0.85485,      0.85586,      0.85686,      0.857
86,      0.85886,      0.85986,      0.86086,      0.86186,      0.86286,      0.8638
6,
      0.86486,      0.86587,      0.86687,      0.86787,      0.86887,      0.8
6987,      0.87087,      0.87187,      0.87287,      0.87387,      0.87487,      0.87
588,      0.87688,      0.87788,      0.87888,      0.87988,      0.88088,      0.881
88,      0.88288,      0.88388,      0.88488,      0.88589,      0.88689,      0.8878
9,
      0.88889,      0.88989,      0.89089,      0.89189,      0.89289,      0.8
9389,      0.89489,      0.8959,      0.8969,      0.8979,      0.8989,      0.8
999,      0.9009,      0.9019,      0.9029,      0.9039,      0.9049,      0.905
91,      0.90691,      0.90791,      0.90891,      0.90991,      0.91091,      0.9119
1,
      0.91291,      0.91391,      0.91491,      0.91592,      0.91692,      0.9
1792,      0.91892,      0.91992,      0.92092,      0.92192,      0.92292,      0.92
392,      0.92492,      0.92593,      0.92693,      0.92793,      0.92893,      0.929
93,      0.93093,      0.93193,      0.93293,      0.93393,      0.93493,      0.9359
4,
      0.93694,      0.93794,      0.93894,      0.93994,      0.94094,      0.9
4194,      0.94294,      0.94394,      0.94494,      0.94595,      0.94695,      0.94
795,      0.94895,      0.94995,      0.95095,      0.95195,      0.95295,      0.953
95,      0.95495,      0.95596,      0.95696,      0.95796,      0.95896,      0.9599
6,
      0.96096,      0.96196,      0.96296,      0.96396,      0.96496,      0.9
6597,      0.96697,      0.96797,      0.96897,      0.96997,      0.97097,      0.97
197,      0.97297,      0.97397,      0.97497,      0.97598,      0.97698,      0.977
98,      0.97898,      0.97998,      0.98098,      0.98198,      0.98298,      0.9839
8,
      0.98498,      0.98599,      0.98699,      0.98799,      0.98899,      0.9
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1])), 'Confidence', 'Precision'], [array([      0,      0.001001,      0.00200
2,      0.003003,      0.004004,      0.005005,      0.006006,      0.007007,      0.00800
8,      0.009009,      0.01001,      0.011011,      0.012012,      0.013013,      0.01401
4,      0.015015,      0.016016,      0.017017,      0.018018,      0.019019,      0.0200
2,      0.021021,      0.022022,      0.023023,
      0.024024,      0.025025,      0.026026,      0.027027,      0.028028,      0.02
9029,      0.03003,      0.031031,      0.032032,      0.033033,      0.034034,      0.035
035,      0.036036,      0.037037,      0.038038,      0.039039,      0.04004,      0.0410
41,      0.042042,      0.043043,      0.044044,      0.045045,      0.046046,      0.04704
7,
      0.048048,      0.049049,      0.05005,      0.051051,      0.052052,      0.05
3053,      0.054054,      0.055055,      0.056056,      0.057057,      0.058058,      0.059
059,      0.06006,      0.061061,      0.062062,      0.063063,      0.064064,      0.0650
65,      0.066066,      0.067067,      0.068068,      0.069069,      0.07007,      0.07107
1,
      0.072072,      0.073073,      0.074074,      0.075075,      0.076076,      0.07

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083,	0.084084,	0.085085,	0.086086,	0.087087,	0.088088,	0.0890
89,	0.09009,	0.091091,	0.092092,	0.093093,	0.094094,	0.09509
5,						
	0.096096,	0.097097,	0.098098,	0.099099,	0.1001,	0.
1011,	0.1021,	0.1031,	0.1041,	0.10511,	0.10611,	0.10
711,	0.10811,	0.10911,	0.11011,	0.11111,	0.11211,	0.113
11,	0.11411,	0.11512,	0.11612,	0.11712,	0.11812,	0.1191
2,						
	0.12012,	0.12112,	0.12212,	0.12312,	0.12412,	0.1
2513,	0.12613,	0.12713,	0.12813,	0.12913,	0.13013,	0.13
113,	0.13213,	0.13313,	0.13413,	0.13514,	0.13614,	0.137
14,	0.13814,	0.13914,	0.14014,	0.14114,	0.14214,	0.1431
4,						
	0.14414,	0.14515,	0.14615,	0.14715,	0.14815,	0.1
4915,	0.15015,	0.15115,	0.15215,	0.15315,	0.15415,	0.15
516,	0.15616,	0.15716,	0.15816,	0.15916,	0.16016,	0.161
16,	0.16216,	0.16316,	0.16416,	0.16517,	0.16617,	0.1671
7,						
	0.16817,	0.16917,	0.17017,	0.17117,	0.17217,	0.1
7317,	0.17417,	0.17518,	0.17618,	0.17718,	0.17818,	0.17
918,	0.18018,	0.18118,	0.18218,	0.18318,	0.18418,	0.185
19,	0.18619,	0.18719,	0.18819,	0.18919,	0.19019,	0.1911
9,						
	0.19219,	0.19319,	0.19419,	0.1952,	0.1962,	0.
1972,	0.1982,	0.1992,	0.2002,	0.2012,	0.2022,	0.2
032,	0.2042,	0.20521,	0.20621,	0.20721,	0.20821,	0.209
21,	0.21021,	0.21121,	0.21221,	0.21321,	0.21421,	0.2152
2,						
	0.21622,	0.21722,	0.21822,	0.21922,	0.22022,	0.2
2122,	0.22222,	0.22322,	0.22422,	0.22523,	0.22623,	0.22
723,	0.22823,	0.22923,	0.23023,	0.23123,	0.23223,	0.233
23,	0.23423,	0.23524,	0.23624,	0.23724,	0.23824,	0.2392
4,						
	0.24024,	0.24124,	0.24224,	0.24324,	0.24424,	0.2
4525,	0.24625,	0.24725,	0.24825,	0.24925,	0.25025,	0.25
125,	0.25225,	0.25325,	0.25425,	0.25526,	0.25626,	0.257
26,	0.25826,	0.25926,	0.26026,	0.26126,	0.26226,	0.2632
6,						
	0.26426,	0.26527,	0.26627,	0.26727,	0.26827,	0.2
6927,	0.27027,	0.27127,	0.27227,	0.27327,	0.27427,	0.27
528,	0.27628,	0.27728,	0.27828,	0.27928,	0.28028,	0.281
28,	0.28228,	0.28328,	0.28428,	0.28529,	0.28629,	0.2872
9,						
	0.28829,	0.28929,	0.29029,	0.29129,	0.29229,	0.2
9329,	0.29429,	0.2953,	0.2963,	0.2973,	0.2983,	0.2
993,	0.3003,	0.3013,	0.3023,	0.3033,	0.3043,	0.305
31,	0.30631,	0.30731,	0.30831,	0.30931,	0.31031,	0.3113
1,						
	0.31231,	0.31331,	0.31431,	0.31532,	0.31632,	0.3
1732,	0.31832,	0.31932,	0.32032,	0.32132,	0.32232,	0.32
332,	0.32432,	0.32533,	0.32633,	0.32733,	0.32833,	0.329
33,	0.33033,	0.33133,	0.33233,	0.33333,	0.33433,	0.3353
4,						
	0.33634,	0.33734,	0.33834,	0.33934,	0.34034,	0.3
4134,	0.34234,	0.34334,	0.34434,	0.34535,	0.34635,	0.34
735,	0.34835,	0.34935,	0.35035,	0.35135,	0.35235,	0.353
35,	0.35435,	0.35536,	0.35636,	0.35736,	0.35836,	0.3593
6,						
	0.36036,	0.36136,	0.36236,	0.36336,	0.36436,	0.3

6537,	0.36637,	0.36737,	0.36837,	0.36937,	0.37037,	0.37
137,	0.37237,	0.37337,	0.37437,	0.37538,	0.37638,	0.377
38,	0.37838,	0.37938,	0.38038,	0.38138,	0.38238,	0.3833
8,						
	0.38438,	0.38539,	0.38639,	0.38739,	0.38839,	0.3
8939,	0.39039,	0.39139,	0.39239,	0.39339,	0.39439,	0.3
954,	0.3964,	0.3974,	0.3984,	0.3994,	0.4004,	0.40
14,	0.4024,	0.4034,	0.4044,	0.40541,	0.40641,	0.4074
1,						
	0.40841,	0.40941,	0.41041,	0.41141,	0.41241,	0.4
1341,	0.41441,	0.41542,	0.41642,	0.41742,	0.41842,	0.41
942,	0.42042,	0.42142,	0.42242,	0.42342,	0.42442,	0.425
43,	0.42643,	0.42743,	0.42843,	0.42943,	0.43043,	0.4314
3,						
	0.43243,	0.43343,	0.43443,	0.43544,	0.43644,	0.4
3744,	0.43844,	0.43944,	0.44044,	0.44144,	0.44244,	0.44
344,	0.44444,	0.44545,	0.44645,	0.44745,	0.44845,	0.449
45,	0.45045,	0.45145,	0.45245,	0.45345,	0.45445,	0.4554
6,						
	0.45646,	0.45746,	0.45846,	0.45946,	0.46046,	0.4
6146,	0.46246,	0.46346,	0.46446,	0.46547,	0.46647,	0.46
747,	0.46847,	0.46947,	0.47047,	0.47147,	0.47247,	0.473
47,	0.47447,	0.47548,	0.47648,	0.47748,	0.47848,	0.4794
8,						
	0.48048,	0.48148,	0.48248,	0.48348,	0.48448,	0.4
8549,	0.48649,	0.48749,	0.48849,	0.48949,	0.49049,	0.49
149,	0.49249,	0.49349,	0.49449,	0.4955,	0.4965,	0.49
75,	0.4985,	0.4995,	0.5005,	0.5015,	0.5025,	0.503
5,						
	0.5045,	0.50551,	0.50651,	0.50751,	0.50851,	0.5
0951,	0.51051,	0.51151,	0.51251,	0.51351,	0.51451,	0.51
552,	0.51652,	0.51752,	0.51852,	0.51952,	0.52052,	0.521
52,	0.52252,	0.52352,	0.52452,	0.52553,	0.52653,	0.5275
3,						
	0.52853,	0.52953,	0.53053,	0.53153,	0.53253,	0.5
3353,	0.53453,	0.53554,	0.53654,	0.53754,	0.53854,	0.53
954,	0.54054,	0.54154,	0.54254,	0.54354,	0.54454,	0.545
55,	0.54655,	0.54755,	0.54855,	0.54955,	0.55055,	0.5515
5,						
	0.55255,	0.55355,	0.55455,	0.55556,	0.55656,	0.5
5756,	0.55856,	0.55956,	0.56056,	0.56156,	0.56256,	0.56
356,	0.56456,	0.56557,	0.56657,	0.56757,	0.56857,	0.569
57,	0.57057,	0.57157,	0.57257,	0.57357,	0.57457,	0.5755
8,						
	0.57658,	0.57758,	0.57858,	0.57958,	0.58058,	0.5
8158,	0.58258,	0.58358,	0.58458,	0.58559,	0.58659,	0.58
759,	0.58859,	0.58959,	0.59059,	0.59159,	0.59259,	0.593
59,	0.59459,	0.5956,	0.5966,	0.5976,	0.5986,	0.599
6,						
	0.6006,	0.6016,	0.6026,	0.6036,	0.6046,	0.6
0561,	0.60661,	0.60761,	0.60861,	0.60961,	0.61061,	0.61
161,	0.61261,	0.61361,	0.61461,	0.61562,	0.61662,	0.617
62,	0.61862,	0.61962,	0.62062,	0.62162,	0.62262,	0.6236
2,						
	0.62462,	0.62563,	0.62663,	0.62763,	0.62863,	0.6
2963,	0.63063,	0.63163,	0.63263,	0.63363,	0.63463,	0.63
564,	0.63664,	0.63764,	0.63864,	0.63964,	0.64064,	0.641
64,	0.64264,	0.64364,	0.64464,	0.64565,	0.64665,	0.6476
5,						
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5365,	0.65465,	0.65566,	0.65666,	0.65766,	0.65866,	0.65
966,	0.66066,	0.66166,	0.66266,	0.66366,	0.66466,	0.665
67,	0.66667,	0.66767,	0.66867,	0.66967,	0.67067,	0.6716
7,						
	0.67267,	0.67367,	0.67467,	0.67568,	0.67668,	0.6
7768,	0.67868,	0.67968,	0.68068,	0.68168,	0.68268,	0.68
368,	0.68468,	0.68569,	0.68669,	0.68769,	0.68869,	0.689
69,	0.69069,	0.69169,	0.69269,	0.69369,	0.69469,	0.695
7,						
	0.6967,	0.6977,	0.6987,	0.6997,	0.7007,	0.
7017,	0.7027,	0.7037,	0.7047,	0.70571,	0.70671,	0.70
771,	0.70871,	0.70971,	0.71071,	0.71171,	0.71271,	0.713
71,	0.71471,	0.71572,	0.71672,	0.71772,	0.71872,	0.7197
2,						
	0.72072,	0.72172,	0.72272,	0.72372,	0.72472,	0.7
2573,	0.72673,	0.72773,	0.72873,	0.72973,	0.73073,	0.73
173,	0.73273,	0.73373,	0.73473,	0.73574,	0.73674,	0.737
74,	0.73874,	0.73974,	0.74074,	0.74174,	0.74274,	0.7437
4,						
	0.74474,	0.74575,	0.74675,	0.74775,	0.74875,	0.7
4975,	0.75075,	0.75175,	0.75275,	0.75375,	0.75475,	0.75
576,	0.75676,	0.75776,	0.75876,	0.75976,	0.76076,	0.761
76,	0.76276,	0.76376,	0.76476,	0.76577,	0.76677,	0.7677
7,						
	0.76877,	0.76977,	0.77077,	0.77177,	0.77277,	0.7
7377,	0.77477,	0.77578,	0.77678,	0.77778,	0.77878,	0.77
978,	0.78078,	0.78178,	0.78278,	0.78378,	0.78478,	0.785
79,	0.78679,	0.78779,	0.78879,	0.78979,	0.79079,	0.7917
9,						
	0.79279,	0.79379,	0.79479,	0.7958,	0.7968,	0.
7978,	0.7988,	0.7998,	0.8008,	0.8018,	0.8028,	0.8
038,	0.8048,	0.80581,	0.80681,	0.80781,	0.80881,	0.809
81,	0.81081,	0.81181,	0.81281,	0.81381,	0.81481,	0.8158
2,						
	0.81682,	0.81782,	0.81882,	0.81982,	0.82082,	0.8
2182,	0.82282,	0.82382,	0.82482,	0.82583,	0.82683,	0.82
783,	0.82883,	0.82983,	0.83083,	0.83183,	0.83283,	0.833
83,	0.83483,	0.83584,	0.83684,	0.83784,	0.83884,	0.8398
4,						
	0.84084,	0.84184,	0.84284,	0.84384,	0.84484,	0.8
4585,	0.84685,	0.84785,	0.84885,	0.84985,	0.85085,	0.85
185,	0.85285,	0.85385,	0.85485,	0.85586,	0.85686,	0.857
86,	0.85886,	0.85986,	0.86086,	0.86186,	0.86286,	0.8638
6,						
	0.86486,	0.86587,	0.86687,	0.86787,	0.86887,	0.8
6987,	0.87087,	0.87187,	0.87287,	0.87387,	0.87487,	0.87
588,	0.87688,	0.87788,	0.87888,	0.87988,	0.88088,	0.881
88,	0.88288,	0.88388,	0.88488,	0.88589,	0.88689,	0.8878
9,						
	0.88889,	0.88989,	0.89089,	0.89189,	0.89289,	0.8
9389,	0.89489,	0.8959,	0.8969,	0.8979,	0.8989,	0.8
999,	0.9009,	0.9019,	0.9029,	0.9039,	0.9049,	0.905
91,	0.90691,	0.90791,	0.90891,	0.90991,	0.91091,	0.9119
1,						
	0.91291,	0.91391,	0.91491,	0.91592,	0.91692,	0.9
1792,	0.91892,	0.91992,	0.92092,	0.92192,	0.92292,	0.92
392,	0.92492,	0.92593,	0.92693,	0.92793,	0.92893,	0.929
93,	0.93093,	0.93193,	0.93293,	0.93393,	0.93493,	0.9359
4,						
	0.93694,	0.93794,	0.93894,	0.93994,	0.94094,	0.9

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4194,    0.94294,    0.94394,    0.94494,    0.94595,    0.94695,    0.94
795,    0.94895,    0.94995,    0.95095,    0.95195,    0.95295,    0.953
95,    0.95495,    0.95596,    0.95696,    0.95796,    0.95896,    0.9599
6,
        0.96096,    0.96196,    0.96296,    0.96396,    0.96496,    0.9
6597,    0.96697,    0.96797,    0.96897,    0.96997,    0.97097,    0.97
197,    0.97297,    0.97397,    0.97497,    0.97598,    0.97698,    0.977
98,    0.97898,    0.97998,    0.98098,    0.98198,    0.98298,    0.9839
8,
        0.98498,    0.98599,    0.98699,    0.98799,    0.98899,    0.9
8999,    0.99099,    0.99199,    0.99299,    0.99399,    0.99499,    0.
996,    0.997,    0.998,    0.999,    1]), array([[ 0.97506,
0.97506,    0.97091, ...,    0,    0,    0],
[ 0.98458,    0.98458,    0.98118, ...,    0,    0,
0],
[ 0.83562,    0.83562,    0.77397, ...,    0,    0,
0]]), 'Confidence', 'Recall']]
fitness: 0.38384026205773014
keys: ['metrics/precision(B)', 'metrics/recall(B)', 'metrics/mAP50(B)', 'metric
s/mAP50-95(B)']
maps: array([ 0.5352,    0.59944,    0.016883])
names: {0: 'head', 1: 'helmet', 2: 'person'}
nt_per_class: array([1203, 3826, 146])
nt_per_image: array([185, 929, 40])
results_dict: {'metrics/precision(B)': 0.931912371252655, 'metrics/recall(B)':
0.5806407019450953, 'metrics/mAP50(B)': 0.6214783810362072, 'metrics/mAP50-95
(B)': 0.38384026205773014, 'fitness': 0.38384026205773014}
save_dir: PosixPath('/content/runs/detect/train')
speed: {'preprocess': 0.23782080200180644, 'inference': 2.251571640002112, 'los
s': 0.0004448140016393154, 'postprocess': 2.4772315879990856}
stats: {'tp': [], 'conf': [], 'pred_cls': [], 'target_cls': [], 'target_img':
[]}]

```

Validation metrics menunjukkan model memiliki precision yang tinggi sebesar 93,19%, sehingga sebagian besar objek yang dideteksi sudah benar. Namun, recall sebesar 58,06% menunjukkan masih ada beberapa objek yang terlewat saat deteksi. Nilai mAP@50 sebesar 62,15% menunjukkan model sudah cukup baik dalam mendeteksi objek, sedangkan mAP@50-95 sebesar 38,38% menunjukkan akurasi bounding box masih perlu ditingkatkan pada evaluasi yang lebih ketat. Berdasarkan mAP per class, class helmet dan head memiliki performa yang cukup baik, sedangkan class person memiliki performa sangat rendah. Hal ini kemungkinan disebabkan jumlah data person yang terlalu sedikit sehingga model kurang mampu mempelajari pola objek tersebut dengan baik.

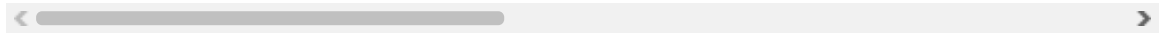
```

In [ ]: df = pd.read_csv('/content/runs/detect/train/results.csv')
df.head()

```

Out[]:

	epoch	time	train/box_loss	train/cls_loss	train/df_l_loss	metrics/precision(B)	me
0	1	109.226	1.65997	2.00205	1.25960	0.87893	
1	2	182.472	1.57449	1.32441	1.23237	0.88583	
2	3	256.614	1.58862	1.22375	1.24238	0.81757	
3	4	328.976	1.55921	1.14207	1.23115	0.87434	
4	5	403.812	1.53764	1.07401	1.22910	0.90074	



Best Epoch Berdasarkan mAP50-95, mAP50, validation box

```
In [ ]: best_epoch = df['metrics/mAP50-95(B)'].idxmax()
best_epoch + 1
```

Out[]: 47

```
In [ ]: best_epoch_map50 = df['metrics/mAP50(B)'].idxmax()
best_epoch_map50
```

Out[]: 39

```
In [ ]: best_epoch_val = df['val/box_loss'].idxmin()
best_epoch_val
```

Out[]: 46

dari hasil training, ketiganya mencapai performa terbaik sebelum epoch 50. Ini menunjukkan bahwa epochs 50 sudah cukup untuk dataset ini, karena model sudah mencapai performanya yang optimal sebelum epoch terakhir. Karena acuan utamanya adalah mAP50-95 (memiliki evaluasi paling ketat), maka epoch 47 yang dijadikan best epoch.

Evaluations

train

```
In [ ]: os.listdir('/content/runs/detect/train')
```

```
Out[ ]: ['val_batch1_pred.jpg',
        'train_batch8761.jpg',
        'train_batch1.jpg',
        'train_batch8760.jpg',
        'labels.jpg',
        'results.png',
        'val_batch1_labels.jpg',
        'BoxR_curve.png',
        'BoxP_curve.png',
        'args.yaml',
        'BoxF1_curve.png',
        'confusion_matrix.png',
        'train_batch8762.jpg',
        'val_batch0_pred.jpg',
        'val_batch2_pred.jpg',
        'train_batch0.jpg',
        'val_batch2_labels.jpg',
        'weights',
        'train_batch2.jpg',
        'results.csv',
        'confusion_matrix_normalized.png',
        'BoxPR_curve.png',
        'val_batch0_labels.jpg']
```

Diatas merupakan output folder hasil training YOLO, yang berisi file-file visualisasi

Evaluation metrics

```
In [ ]: import matplotlib.image as mpimg

model = YOLO('/content/runs/detect/train/weights/best.pt')

# evaluasi test set
model.val(
    data='/content/Hard-Hat-Detection-1/data.yaml',
    split='test',
    project='/content/evaluations',
    name='test_eval_v1',
    exist_ok=True
)

os.listdir('/content/evaluations/test_eval_v1')
```

Ultralytics 8.4.49 🚀 Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
 YOLO11n summary (fused): 101 layers, 2,582,737 parameters, 0 gradients, 6.3 GFLOPs
 val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 1378.3±264.3 MB/s, size: 49.2 KB)
 val: Scanning /content/Hard-Hat-Detection-1/test/labels.cache... 500 images, 0 backgrounds, 0 corrupt: 100% ————— 500/500 123.4Mit/s 0.0s

	Class	Images	Instances	Box(P	R	mAP50	mA
P50-95): 100%		32/32	4.3it/s	7.5s			
	all	500	2686	0.939	0.581	0.629	
	head	99	764	0.907	0.855	0.918	
	helmet	446	1857	0.91	0.889	0.939	
	person	13	65	1	0	0.0284	

Speed: 2.7ms preprocess, 4.1ms inference, 0.0ms loss, 1.7ms postprocess per image
 Results saved to /content/evaluations/test_eval_v1

```
Out[ ]: ['val_batch1_pred.jpg',
        'val_batch1_labels.jpg',
        'BoxR_curve.png',
        'BoxP_curve.png',
        'BoxF1_curve.png',
        'confusion_matrix.png',
        'val_batch0_pred.jpg',
        'val_batch2_pred.jpg',
        'val_batch2_labels.jpg',
        'confusion_matrix_normalized.png',
        'BoxPR_curve.png',
        'val_batch0_labels.jpg']
```

- Evaluasi overall : Precision sebesar 93,9% menunjukkan bahwa sebagian besar objek yang dideteksi model sudah benar. Jadi, model jarang salah mendeteksi objek yang sebenarnya tidak ada (false positive rendah). Recall sebesar 58,1% menunjukkan bahwa model masih melewatkan cukup banyak objek yang ada pada gambar. Artinya, kemampuan model dalam menemukan seluruh objek belum optimal, terutama untuk objek yang kecil, kurang terlihat jelas, atau kurang jelas. Nilai mAP@50 sebesar 62,9% menunjukkan bahwa model cukup mampu mendeteksi objek dan menghasilkan bounding box yang cukup sesuai dengan posisi objek sebenarnya. Namun, nilai mAP@50-95 yang hanya sebesar 38,7% menunjukkan bahwa ketika evaluasi dilakukan dengan standar yang lebih ketat, performa model masih menurun. Hal ini menandakan posisi bounding box yang dihasilkan model masih belum terlalu presisi. Secara keseluruhan, performa model belum optimal yang diakibatkan oleh distribusi data training yang tidak seimbang.
- Evaluasi per-class : Helmet performanya paling bagus, precision 0.91, recall 0.889, mAP50-95 0.593 karena datanya paling banyak. Head juga cukup baik, recall 0.855 dan mAP50-95 0.552. Sedangkan person sangat buruk, recall 0 dan mAP50-95 hanya 0.0165 karena datanya hanya ada 800 instance.
 Ini menjadi bukti yang kuat bahwa penyebab utama recall dan map rendah secara keseluruhan memang kelas person. Model hampir tidak bisa mendeteksi person

sama sekali, yang akhirnya langsung menarik turun nilai rata-rata keseluruhan. Kalau model ini mau ditingkatkan, fokus perbaikannya sebaiknya di kelas person, misalnya dengan menambah variasi data atau memperbaiki distribusi dataset agar model bisa belajar lebih seimbang.

Grafik Train Loss, Validation Loss, dan Evaluation Metrics

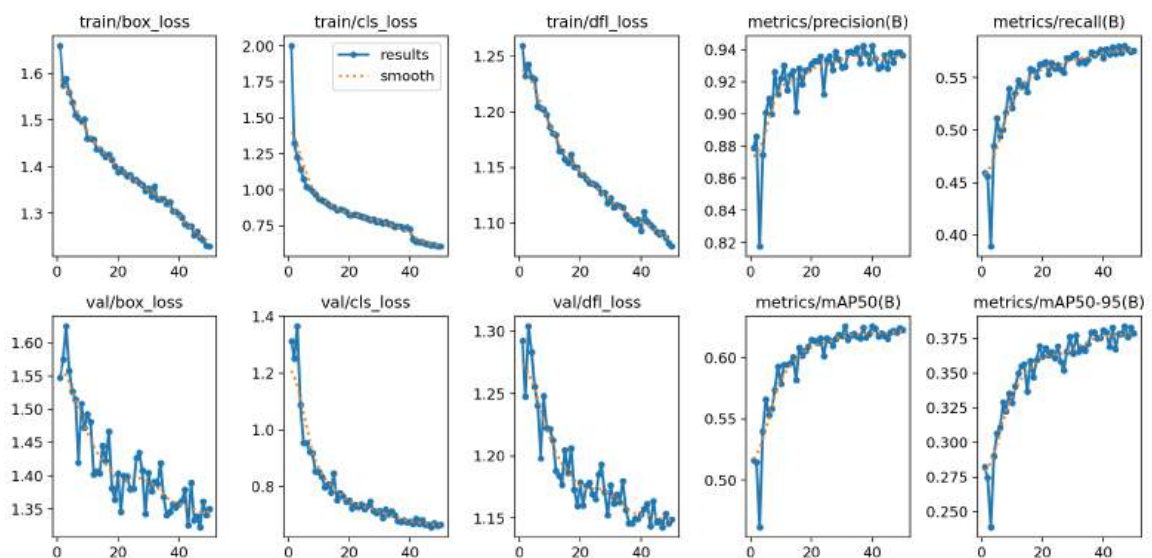
YOLO memisahkan loss-nya jadi 3 komponen karena masing-masing mengukur hal yang berbeda:

- Box loss : mengukur seberapa tepat posisi dan ukuran bounding box prediksi dibanding ground truth
- Cls loss : mengukur seberapa tepat klasifikasi kelasnya (head, helmet, person)
- Dfl loss : mengukur seberapa presisi distribusi lokasi tepi bounding box-nya(lebih detail dari box loss)

```
In [ ]: from IPython.display import Image
```

```
Image('/content/runs/detect/train/results.png')
```

```
Out[ ]:
```



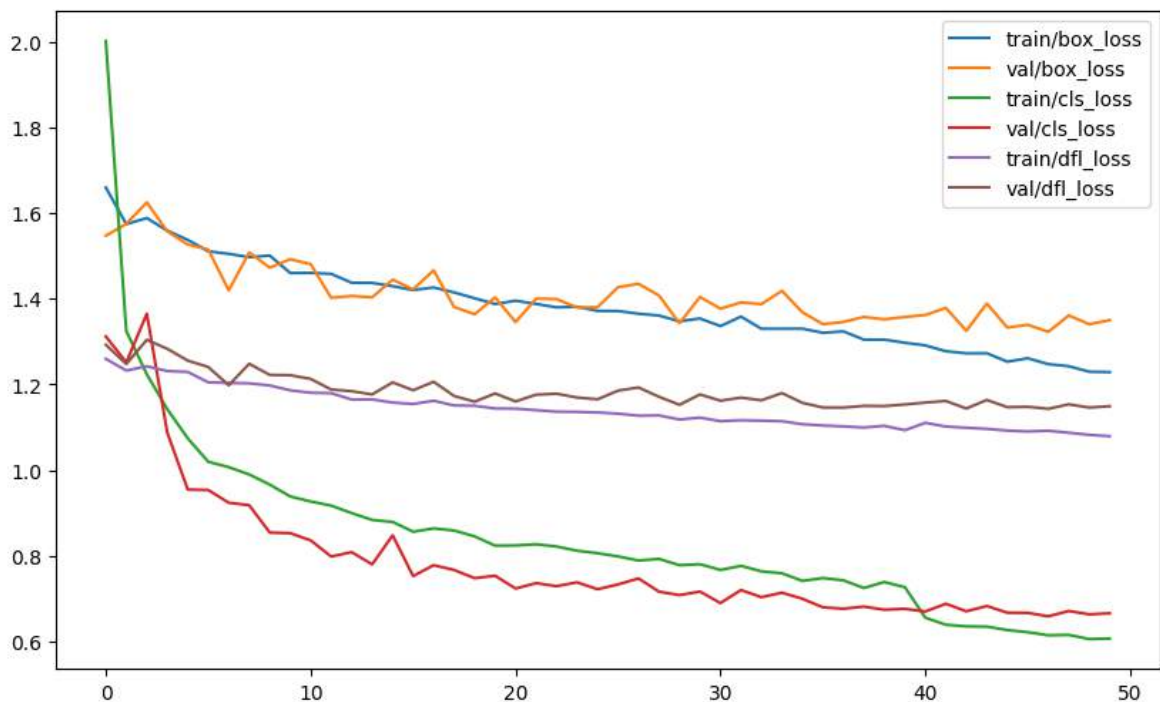
- Loss : train dan val loss keduanya turun konsisten sampai akhir, mengindikasikan proses training berjalan stabil.
- Precision : naik cepat di awal dan stabil di sekitar 0.93 mulai epoch 20-an, artinya model cepat belajar membuat prediksi yang akurat.
- Recall : naik lebih lambat dan fluktuatif, mencerminkan model kesulitan mendeteksi semua objek, konsisten dengan masalah class imbalance pada kelas person.
- mAP50 & mAP50-95 — keduanya naik dan mulai stabil mendekati epoch 47, sesuai dengan best epoch yang ditemukan sebelumnya, menandakan model sudah mencapai performa optimalnya di situ.

```
In [ ]: df = pd.read_csv('/content/runs/detect/train/results.csv')
```

```
df[['train/box_loss', 'val/box_loss',
```

```
'train/cls_loss', 'val/cls_loss',  
'train/df1_loss', 'val/df1_loss']].plot(figsize=(10,6))
```

Out[]: <Axes: >



Dari grafik train dan validation loss ini, model cenderung menunjukkan kondisi good fit menuju mild overfitting, tapi belum termasuk overfit parah.

Hal ini terlihat karena train loss dan validation loss sama-sama menurun selama training. Jadi model masih belajar dengan baik pada data training maupun validation. Selain itu, pola grafik train dan val juga masih cukup searah dan tidak menunjukkan divergence yang ekstrem.

Namun, tetap ada gap antara train loss dan validation loss, terutama pada box loss dan df1 loss. Train loss terus turun, sementara validation loss turun lebih lambat dan cenderung stagnan di akhir epoch. Ini menunjukkan mulai ada indikasi overfitting ringan karena model belajar lebih baik pada data training dibanding validation.

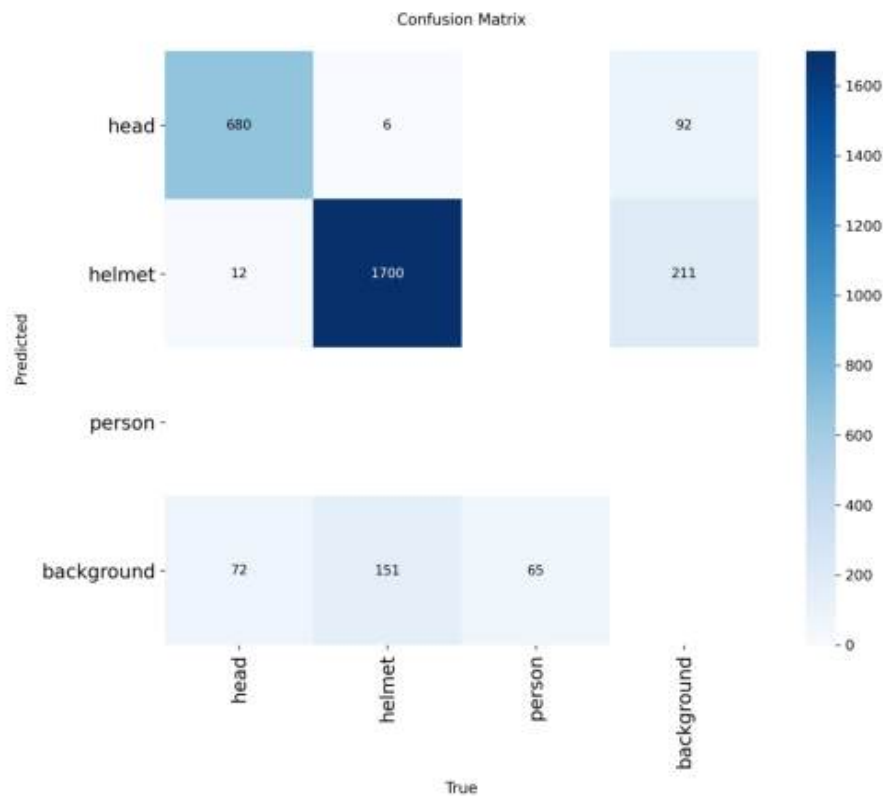
Besarnya gap antar train dan validation loss masih tergolong tidak terlalu besar. Selisihnya rata-rata sekitar 0.1–0.2, sehingga masih bisa dianggap mild overfitting. Jika gap sampai mendekati 1 atau validation loss terus naik jauh saat train loss turun, barulah bisa dianggap overfit berat.

Untuk konvergensi, model terlihat sudah hampir konvergen di akhir training. Ini terlihat dari grafik loss yang mulai stabil dan penurunannya semakin kecil setelah sekitar epoch 35–40. Artinya, model sudah mendekati performa optimalnya dan tambahan epoch kemungkinan tidak akan memberikan peningkatan yang signifikan.

Confusion Matrix

```
In [ ]: img = mpimg.imread('/content/evaluations/test_eval_v1/confusion_matrix.png')
```

```
plt.figure(figsize=(8,8))
plt.imshow(img)
plt.axis('off')
plt.show()
```



Hasil confusion matrix :

- Head : dari 778 instance (680+6+92), 680 berhasil terdeteksi benar. Yang 92 missed (tidak terdeteksi sama sekali) dan 6 salah diklasifikasi sebagai helmet (kesalahan predict nya hanya sekitar 14%). Performanya cukup baik.
- Helmet : dari 1923 instance (12+1700+211), 1700 terdeteksi benar. Yang 211 missed dan 12 salah diklasifikasi sebagai head (kesalahan predict nya juga hanya sekitar 13%). Hasilnya juga cukup baik.
- Person : semua instance tidak terdeteksi sama sekali (seluruhnya masuk ke background), bukti bahwa model benar-benar gagal mendeteksi kelas ini akibat datanya yang sangat sedikit.
- Background : ada 72 head, 151 helmet, dan 65 person yang diprediksi sebagai background, artinya model cukup sering melewati objek terutama pada kelas person. Kesimpulannya confusion matrix ini memperkuat argumen kami sebelumnya bahwa kelas person adalah masalah utama, dan solusinya adalah hapus kelas person, atau kalau tetap ingin dipertahankan baru tambah data secara signifikan.

Test Precision-Recall Curve

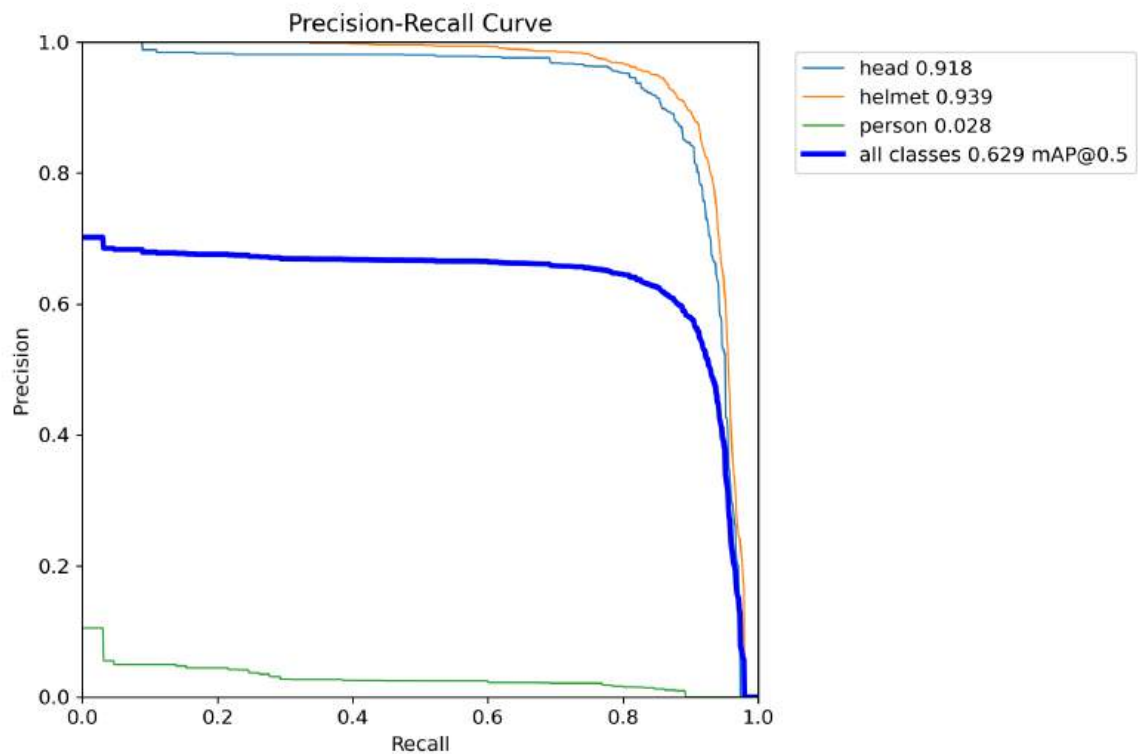
Biru muda : head

Orange : helmet

Hijau : person

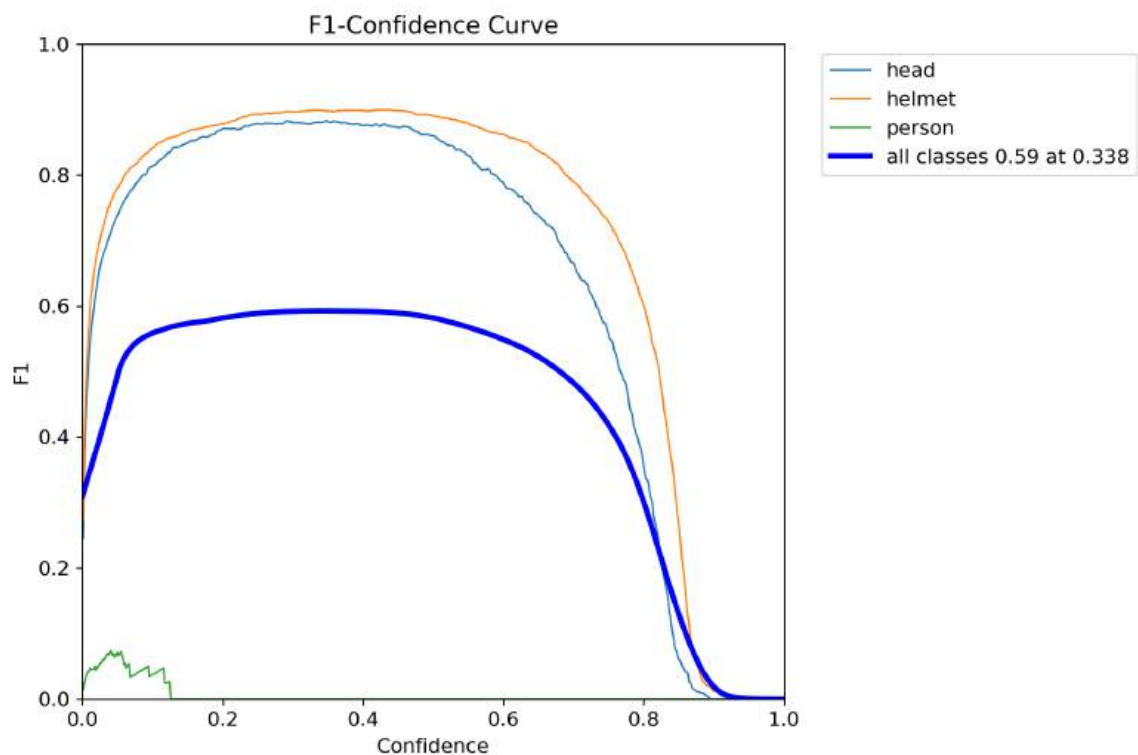
Biru tua (tebal) : all classes (rata-rata keseluruhan)

```
In [ ]: display(Image('/content/evaluations/test_eval_v1/BoxPR_curve.png'))
```



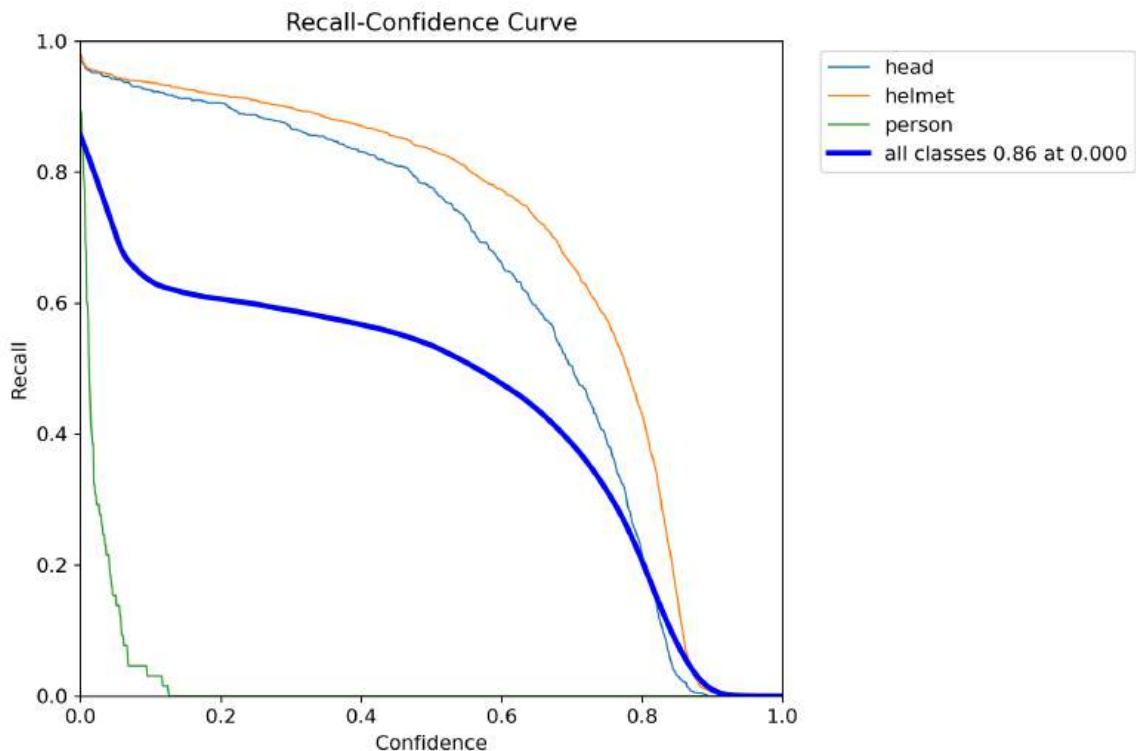
Test F1 Curve

```
In [ ]: display(Image('/content/evaluations/test_eval_v1/BoxF1_curve.png'))
```



Test Recall Curve

```
In [ ]: display(Image('/content/evaluations/test_eval_v1/BoxR_curve.png'))
```



- Precision-Recall Curve : head (0.918) dan helmet (0.939) kurva nya tinggi dan lebar, artinya keduanya performa bagus di berbagai threshold. Person (0.028) hampir flat di bawah, konfirmasi model gagal total mendeteksi kelas ini.
- F1-Confidence Curve : F1 score terbaik untuk semua kelas ada di confidence 0.338. Artinya threshold optimal model bukan di 0.5 (default), tapi lebih rendah. Kalau pakai threshold 0.5, performa model sebenarnya lebih buruk dari yang seharusnya. Head dan helmet F1-nya bagus di sekitar 0.87-0.89, sedangkan person hampir 0 di semua confidence.
- Recall-Confidence Curve : recall head dan helmet tinggi di confidence rendah tapi turun seiring confidence naik, yang wajar. Person langsung drop ke 0 bahkan di confidence sangat rendah, artinya model benar-benar tidak mengenali kelas ini sama sekali.

Ketiga grafik ini secara konsisten memperkuat kesimpulan yang sama, yaitu head dan helmet sudah cukup baik. Person adalah masalah utama akibat class imbalance.

Sample Test Detection Results

Gunanya untuk melihat visualisasi hasil

```
In [ ]: results = model.predict(
    source='/content/Hard-Hat-Detection-1/test/images',
    save=True,
```

```
) conf=0.25
```

image 1/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1006_png.r
f.256a979b0b85f9d8b2a68bd16af7e51b.jpg: 640x640 17 heads, 2 helmets, 12.0ms
image 2/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1018_png.r
f.c50cb2c577064fbbba7d47fb649da56ae.jpg: 640x640 4 helmets, 8.3ms
image 3/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1019_png.r
f.9222efe332c6efb697531806a0c11125.jpg: 640x640 4 helmets, 8.7ms
image 4/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1020_png.r
f.baa8a9818b49f4c547ed8ce1f30511cd.jpg: 640x640 8 helmets, 7.8ms
image 5/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1022_png.r
f.9160c68da3e2854d01493902e1669fa0.jpg: 640x640 4 helmets, 7.8ms
image 6/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1023_png.r
f.1628f4d867cfadb3564d456526e1ad7f.jpg: 640x640 8 helmets, 8.5ms
image 7/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1025_png.r
f.65552c36a3a31e7638d84a10abf97d9f.jpg: 640x640 3 helmets, 7.9ms
image 8/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1027_png.r
f.63824ed89c255beea28eeca1709613a9.jpg: 640x640 4 helmets, 8.2ms
image 9/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1034_png.r
f.434d7159df9d34f5cfb6b9efa52185b6.jpg: 640x640 12 heads, 7.8ms
image 10/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1042_png.r
f.7173ff4d646599ca2421b970b03109b5.jpg: 640x640 1 head, 1 helmet, 8.1ms
image 11/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1053_png.r
f.01c1a98d2e9e0559c479d69b26ef7483.jpg: 640x640 4 helmets, 7.8ms
image 12/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1055_png.r
f.44857927120673080758fa34629ed7b2.jpg: 640x640 1 head, 1 helmet, 7.8ms
image 13/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1057_png.r
f.feb157ce02531ff7b16e1de22de6e746.jpg: 640x640 4 helmets, 7.8ms
image 14/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1065_png.r
f.c4451cd976ce515a3e4e70c51c950a94.jpg: 640x640 1 helmet, 7.8ms
image 15/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1082_png.r
f.eddb8220718698771a0175140d85676b.jpg: 640x640 2 heads, 13 helmets, 7.8ms
image 16/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1087_png.r
f.29370f927323e4dd25d6bc3e9447a1df.jpg: 640x640 1 head, 1 helmet, 9.8ms
image 17/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1108_png.r
f.f27351f9376018510dd5730cf53f4ef5.jpg: 640x640 4 heads, 2 helmets, 7.9ms
image 18/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1116_png.r
f.87cbf7ea3b8a1130e659b1c2bdec3411.jpg: 640x640 3 helmets, 8.1ms
image 19/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1140_png.r
f.de76beea312f5199290dea71d2a7104b.jpg: 640x640 6 helmets, 7.9ms
image 20/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1144_png.r
f.e2628d729b4e77d9252b80503ca664f6.jpg: 640x640 2 helmets, 7.9ms
image 21/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1152_png.r
f.6ed95b3047c1ebe370c77dbadf2d6569.jpg: 640x640 2 helmets, 8.0ms
image 22/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1169_png.r
f.559ae3d87e1cf3c948bdf3692cd7bf09.jpg: 640x640 1 helmet, 7.8ms
image 23/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1174_png.r
f.28965107ef1c89e8bba51679a32c20eb.jpg: 640x640 8 heads, 7 helmets, 8.3ms
image 24/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1175_png.r
f.2da52b01bae1f461bba1b75ea2695433.jpg: 640x640 3 helmets, 7.4ms
image 25/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1180_png.r
f.472de3329b7fe9c392f872d0cf71db8d.jpg: 640x640 3 helmets, 7.3ms
image 26/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1190_png.r
f.e146d500acbe4433c9beb6ea6540db33.jpg: 640x640 2 helmets, 8.3ms
image 27/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1198_png.r
f.0109e6bcff3d6e1f954d6f0f72560bd8.jpg: 640x640 1 helmet, 7.6ms
image 28/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1201_png.r
f.13d12e02a4e45d1e124309e222e6f92d.jpg: 640x640 1 helmet, 10.2ms
image 29/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1225_png.r
f.e86ec609c275dfce96613bd82b0d2f5b.jpg: 640x640 12 helmets, 13.7ms
image 30/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1228_png.r
f.aab6747b5272084240e8fdbcb8d1cf2c.jpg: 640x640 2 helmets, 7.4ms

image 31/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1238_png.r
f.7c0763ead0b43cf7515bc7d6738844ff.jpg: 640x640 7 helmets, 8.8ms
image 32/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1242_png.r
f.6d1acb611ef35b864ef4e26370d8f6fc.jpg: 640x640 3 helmets, 7.4ms
image 33/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1263_png.r
f.6417c6ff93d620d5f8da26388ba0d49b.jpg: 640x640 3 helmets, 7.9ms
image 34/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1264_png.r
f.acbac8a98c8f304101c8ba5e71851c02.jpg: 640x640 10 heads, 9.5ms
image 35/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1266_png.r
f.0652d67fd8850ce238fc9fccb4f4f5fb.jpg: 640x640 4 helmets, 7.3ms
image 36/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1270_png.r
f.d8d3ebf85cf9754bbcb2cb08726a1c003.jpg: 640x640 1 head, 4 helmets, 7.3ms
image 37/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1288_png.r
f.2b21f6ef3283d54c5555c53e18183498.jpg: 640x640 9 helmets, 8.1ms
image 38/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1307_png.r
f.d2ad0b7d77515ca795c2a94ae103fe8a.jpg: 640x640 3 helmets, 7.2ms
image 39/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1313_png.r
f.bf79b6178658ca125659e50617d47f8a.jpg: 640x640 29 helmets, 6.8ms
image 40/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1318_png.r
f.e0a6e634578adbcbf318eb6c7b8a1498f.jpg: 640x640 2 helmets, 7.5ms
image 41/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1324_png.r
f.0cf3b140ff604c82f752d219ab4561ea.jpg: 640x640 12 helmets, 9.7ms
image 42/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1351_png.r
f.36b326644ca39be1426c3fe2d36d0d4d.jpg: 640x640 4 helmets, 7.6ms
image 43/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1362_png.r
f.5d340029d0b8de5dc1241b8a88eeff96.jpg: 640x640 9 helmets, 7.3ms
image 44/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1363_png.r
f.3e6d01c75cbbd2a366a1b5ccbec176ce.jpg: 640x640 2 helmets, 7.7ms
image 45/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1364_png.r
f.f01d44be3bed5a290d3b9d7fffb9b3d8.jpg: 640x640 1 helmet, 7.1ms
image 46/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1370_png.r
f.a12b141d35c3deb346528009acafcfa5.jpg: 640x640 3 helmets, 8.6ms
image 47/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1372_png.r
f.8f60351f3f8a1c7383d2bb8e071e0389.jpg: 640x640 3 helmets, 7.6ms
image 48/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1379_png.r
f.07b7bfb6c80e41de75567f897e3fc9f9.jpg: 640x640 14 helmets, 7.3ms
image 49/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1381_png.r
f.05069ad923dff19f4148afbacc0f95b7.jpg: 640x640 2 helmets, 7.0ms
image 50/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1388_png.r
f.e47383f044da6b4d1a8f01c9fd8f5f77.jpg: 640x640 13 heads, 8.2ms
image 51/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1390_png.r
f.38561ae6039363f513ace2033e464c8e.jpg: 640x640 3 helmets, 8.3ms
image 52/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1399_png.r
f.73e4ad9e918a4f2742cd98cf73613255.jpg: 640x640 1 helmet, 7.2ms
image 53/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers139_png.r
f.94a3d57724d1af8da4cc7940083134b8.jpg: 640x640 6 helmets, 7.4ms
image 54/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1400_png.r
f.19fdbf0a4173aa200d562e3159174c0c.jpg: 640x640 7 helmets, 8.5ms
image 55/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1401_png.r
f.8d10127bc9c9c6fb2543faf5f2869ebc.jpg: 640x640 6 helmets, 14.2ms
image 56/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1418_png.r
f.5316866fb0acb5d1cb106d1abc719582.jpg: 640x640 8 helmets, 23.9ms
image 57/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1420_png.r
f.69121183e73a177f3917a2a6817a0fbc.jpg: 640x640 4 helmets, 13.6ms
image 58/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1428_png.r
f.f13f07a59d439faa8e27ceeddb26eb39.jpg: 640x640 7 helmets, 15.6ms
image 59/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1433_png.r
f.27466ba0c7b8a128d3c7334e8f194ece.jpg: 640x640 1 helmet, 7.2ms
image 60/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1437_png.r
f.964578764b4a670891d7a809c66a223e.jpg: 640x640 2 heads, 13 helmets, 7.3ms

image 61/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1438_png.r
f.7f7a15db5e411e19f3e47beaed057468.jpg: 640x640 5 helmets, 7.5ms
image 62/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1457_png.r
f.92c3f5d8502f61e8b360bd3d6b683843.jpg: 640x640 2 helmets, 7.2ms
image 63/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1459_png.r
f.c64ff9cb3bb8b733cf0c47501b50be4c.jpg: 640x640 3 helmets, 7.0ms
image 64/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1465_png.r
f.44e6fd2709a1da5f67adfab8d7aea0c9.jpg: 640x640 2 helmets, 7.2ms
image 65/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1486_png.r
f.25ba94b649a37dfd9842bf85fa4a7938.jpg: 640x640 9 heads, 2 helmets, 8.1ms
image 66/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1491_png.r
f.ea7d0dedcd4abb57b8948f39df472cf2.jpg: 640x640 8 helmets, 7.0ms
image 67/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1496_png.r
f.e85ca1f179fc0071e328918fed908014.jpg: 640x640 4 helmets, 7.3ms
image 68/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1526_png.r
f.24e068db44737a364cf005786803102b.jpg: 640x640 1 head, 8 helmets, 9.0ms
image 69/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1527_png.r
f.36bf97ccd90db3568dc8c565edeca978.jpg: 640x640 2 helmets, 7.3ms
image 70/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers154_png.r
f.a5719dfffa08e8bbc70bb516542538152.jpg: 640x640 5 helmets, 8.5ms
image 71/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1557_png.r
f.ae490ba7256464045dd282253bd9e731.jpg: 640x640 8 helmets, 18.2ms
image 72/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers156_png.r
f.29accd4dfe6d2715d6b8ce6ab12dc2da.jpg: 640x640 2 helmets, 13.2ms
image 73/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1573_png.r
f.6dd9fe32c518a265a2c8784c723be5ad.jpg: 640x640 6 helmets, 13.6ms
image 74/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1580_png.r
f.1512d446f1993a422f8ea217ae7b5bed.jpg: 640x640 1 head, 3 helmets, 13.7ms
image 75/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1583_png.r
f.e4df27db235ba7bc30472f313e3abc9c.jpg: 640x640 2 helmets, 13.5ms
image 76/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1598_png.r
f.ce1986d8be884dbc567ff9142a544d5e.jpg: 640x640 1 head, 10 helmets, 16.7ms
image 77/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1601_png.r
f.54890a0b22a1cf31b9f700a37ad80907.jpg: 640x640 6 helmets, 17.0ms
image 78/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1616_png.r
f.5a94e59edf0e733cbdf462b31c5ac2b3.jpg: 640x640 7 helmets, 24.9ms
image 79/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1623_png.r
f.1349c680642f6bd5594e63d489ab0cb0.jpg: 640x640 9 helmets, 15.9ms
image 80/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1631_png.r
f.1d14397e95bc20b982aee304273498ad.jpg: 640x640 4 helmets, 17.0ms
image 81/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1638_png.r
f.e40d0a673fde685b44862ee2681ecc15.jpg: 640x640 2 helmets, 15.9ms
image 82/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1641_png.r
f.bc4fd9f14ead2f0934628389f9077996.jpg: 640x640 10 heads, 1 helmet, 8.7ms
image 83/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1645_png.r
f.371c998df14c557d009f125556fa64d5.jpg: 640x640 2 helmets, 7.4ms
image 84/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1656_png.r
f.e64e003c70beb04ef25b441dc95883f0.jpg: 640x640 19 helmets, 7.8ms
image 85/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1670_png.r
f.466097eade8a3dd9c1087bd4eec6e681.jpg: 640x640 11 heads, 7.5ms
image 86/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1675_png.r
f.54f0291759be46e4e928a0463c345e09.jpg: 640x640 1 helmet, 7.5ms
image 87/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1707_png.r
f.137c829f33e51ba25c27e742646053b8.jpg: 640x640 4 helmets, 7.6ms
image 88/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1717_png.r
f.28ac2fc412dcf95f63818a7fddec5bcb.jpg: 640x640 10 helmets, 7.5ms
image 89/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1721_png.r
f.39fd5682e832ff069dce589f416a32bb.jpg: 640x640 1 helmet, 8.4ms
image 90/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1723_png.r
f.22578d358eda887e6dff001d57caec58.jpg: 640x640 1 helmet, 7.4ms

image 91/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1742_png.rf.97f851266c48db562efaf528fa15608e.jpg: 640x640 5 helmets, 7.4ms
image 92/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1751_png.rf.2df273350728c852b7ad5b41d1030f60.jpg: 640x640 16 heads, 7.0ms
image 93/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1755_png.rf.12cd2e196e801c0b4e6406d9a0c51394.jpg: 640x640 14 heads, 13.2ms
image 94/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1758_png.rf.f4ee4db037d820e3025717f1f500d0fc.jpg: 640x640 9 helmets, 7.3ms
image 95/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1763_png.rf.c1996bb7724f51ba4dda76b9d0288b8e.jpg: 640x640 6 heads, 30.2ms
image 96/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1766_png.rf.dbf1631b8976e7b510f39e35395463cb.jpg: 640x640 1 head, 9 helmets, 8.8ms
image 97/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1773_png.rf.fcd2ce199d9c3fdef95468f135942ef1.jpg: 640x640 7 heads, 8.2ms
image 98/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1777_png.rf.ec3cba38eaf907d3ec5f3113912ab202.jpg: 640x640 8 heads, 8.2ms
image 99/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1781_png.rf.b8005295aa1c0359842bf2c6479fdb30.jpg: 640x640 2 helmets, 7.7ms
image 100/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers178_png.rf.36491840e859946840d7ac77b548c0a7.jpg: 640x640 3 helmets, 7.5ms
image 101/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1802_png.rf.e5f69a7523363730adc21e407a6b3055.jpg: 640x640 2 helmets, 8.9ms
image 102/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1806_png.rf.243341caaef0fed46b7a18aab2c05f143.jpg: 640x640 3 helmets, 7.2ms
image 103/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers182_png.rf.e7cc34978f91e55ae0c8a97a46899b33.jpg: 640x640 4 helmets, 7.6ms
image 104/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1831_png.rf.25229db87f3162578662be5d98a27823.jpg: 640x640 5 helmets, 7.8ms
image 105/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers183_png.rf.94800e9703fb178268ad8f4be9f1c7f8.jpg: 640x640 2 helmets, 12.7ms
image 106/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1851_png.rf.b67a73a2e5775016340f1691e92440d8.jpg: 640x640 1 helmet, 7.3ms
image 107/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1854_png.rf.3edbd015302b9bd0546dee5c5644cdd7.jpg: 640x640 2 helmets, 8.0ms
image 108/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1879_png.rf.f0c1000809eedd4a48b88ca5183cf828.jpg: 640x640 4 heads, 14 helmets, 7.6ms
image 109/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1880_png.rf.71ab48e0fb08a8d14bba0a0e3d01ca80.jpg: 640x640 2 helmets, 7.6ms
image 110/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1908_png.rf.bb163e077fab1bb2f1eede1fa0ace6d0.jpg: 640x640 3 helmets, 7.2ms
image 111/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers191_png.rf.6a23fe639b8fc75c2191125d093e3149.jpg: 640x640 6 helmets, 7.4ms
image 112/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1921_png.rf.e0b7ae48dead2f9c3e13cdcaf8e4a02e.jpg: 640x640 3 helmets, 8.9ms
image 113/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1929_png.rf.d34492e71590c38b9317ad954bf5a5a1.jpg: 640x640 3 helmets, 8.5ms
image 114/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1936_png.rf.9d326b13ac91d0d7d184d3b51068c617.jpg: 640x640 1 helmet, 7.5ms
image 115/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1942_png.rf.b9f3ed1b5711c51203f7ce42aeb3f967.jpg: 640x640 3 helmets, 7.4ms
image 116/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1952_png.rf.ed40ae2dd94de69728d194f9f54a2471.jpg: 640x640 6 helmets, 7.4ms
image 117/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1963_png.rf.2c0df098f518c0dde4eecefc4c91047a.jpg: 640x640 5 heads, 7.6ms
image 118/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1968_png.rf.2b0819d15674960fdac545c3dce89aad.jpg: 640x640 4 helmets, 7.4ms
image 119/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1977_png.rf.cb4c04bbd8355535620bbf2a9ccbb5f.jpg: 640x640 3 helmets, 8.5ms
image 120/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1997_png.rf.fcb5d9eec41572b16d89d37644f97ebf.jpg: 640x640 3 helmets, 8.4ms

image 121/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers1998_png.rf.888143775820470a061be571b9eb4315.jpg: 640x640 1 helmet, 7.9ms
image 122/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2000_png.rf.3e961d7e0e17bae1367bf22779d2429e.jpg: 640x640 10 helmets, 7.2ms
image 123/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2005_png.rf.6f9902ef219755261ce499119f7f2b73.jpg: 640x640 2 helmets, 8.2ms
image 124/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2009_png.rf.692328db95000ae0aeba6f4cc3beafdb.jpg: 640x640 1 helmet, 7.7ms
image 125/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2013_png.rf.30557a9183406c0788b4369f710b9109.jpg: 640x640 6 helmets, 8.0ms
image 126/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2021_png.rf.eb45b7a25a0763d35e6b3916ff4712f9.jpg: 640x640 5 helmets, 7.5ms
image 127/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2024_png.rf.3fe046a3e763f76dd7fcb3f72af01081.jpg: 640x640 2 helmets, 8.0ms
image 128/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2026_png.rf.50f7772946755e1c004d2f78dfec41e6.jpg: 640x640 8 helmets, 19.3ms
image 129/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2029_png.rf.89cbcacda885eee2993fdc5714319437.jpg: 640x640 5 helmets, 21.3ms
image 130/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2030_png.rf.298a82cbec74724fa60ae4884480589b.jpg: 640x640 1 helmet, 7.7ms
image 131/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2038_png.rf.7998f833147170d185da67b9cadeb374.jpg: 640x640 5 helmets, 7.4ms
image 132/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2049_png.rf.7780bcc152c62723cb255b40240f2bd3.jpg: 640x640 3 helmets, 8.9ms
image 133/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2057_png.rf.22d51aa1138673e67b98cff6ba387be4.jpg: 640x640 2 helmets, 8.2ms
image 134/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2062_png.rf.b8657f6c2f3b5f539bec89ded985973a.jpg: 640x640 2 helmets, 8.5ms
image 135/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2067_png.rf.9740f2de0e6d4f681bebe65d2c1cb3dc.jpg: 640x640 4 helmets, 7.5ms
image 136/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2081_png.rf.bcfec2b5e97e0409e3ef85afe01ce89c.jpg: 640x640 5 helmets, 7.6ms
image 137/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2092_png.rf.08d75429f4369cdedae83fad33b785f.jpg: 640x640 1 head, 2 helmets, 7.7ms
image 138/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2107_png.rf.7f942a932811dd3b270dbd1ef1b02cb5.jpg: 640x640 1 helmet, 7.5ms
image 139/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2132_png.rf.1c676b0a52e53d9be2af9426eddc7826.jpg: 640x640 8 helmets, 8.1ms
image 140/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2142_png.rf.edd553235761e53159d524f4424cddde.jpg: 640x640 8 helmets, 7.4ms
image 141/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2167_png.rf.3f0b2c627e3f08ed7c30aa6a9047acd6.jpg: 640x640 4 helmets, 7.5ms
image 142/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2173_png.rf.a3a7e1f597e55753eac32da2d81061f3.jpg: 640x640 5 heads, 11.7ms
image 143/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2178_png.rf.6358c03c5e2fe4fa51ad9430d60fc2c9.jpg: 640x640 1 helmet, 7.6ms
image 144/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2200_png.rf.046b36f32bdde9f3033a8601e6332c99.jpg: 640x640 10 helmets, 7.3ms
image 145/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2202_png.rf.de43f7c01354b122c8e8fad6ad6e6f4a.jpg: 640x640 4 helmets, 9.2ms
image 146/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2214_png.rf.e7bad57ce4ffb9ee12b107b16c836194.jpg: 640x640 2 helmets, 11.8ms
image 147/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2217_png.rf.9f8c64fd42324742b9c17124c1a3e7b9.jpg: 640x640 4 helmets, 7.6ms
image 148/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2225_png.rf.5e2ff15b782f24ed288c0602bbfe609b.jpg: 640x640 2 helmets, 7.5ms
image 149/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2241_png.rf.66597ac63595d57c46b9505295c6ab94.jpg: 640x640 2 helmets, 7.6ms
image 150/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2242_png.rf.11fa22a3fbc661d5ebc2ae26602dd41b.jpg: 640x640 2 helmets, 7.5ms

image 151/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2243_png.rf.aa68b3a1eae661eef525d6fe8cb1c180.jpg: 640x640 26 heads, 11.4ms
image 152/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2255_png.rf.2f28a2084251b72666da17ab67866fd7.jpg: 640x640 6 helmets, 8.3ms
image 153/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2280_png.rf.daf255ea93ce323c012ede185aacee23.jpg: 640x640 6 helmets, 8.5ms
image 154/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2289_png.rf.b8386c74eadd0d4d378ade09f3f358c1.jpg: 640x640 27 heads, 1 helmet, 7.8ms
image 155/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2297_png.rf.58a6debc38a30b567f9759d54d02d357.jpg: 640x640 9 helmets, 7.6ms
image 156/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2307_png.rf.22d695142a9b187ef338fde937ab9d29.jpg: 640x640 2 helmets, 7.6ms
image 157/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2312_png.rf.717eb93cf11fbc406913e7bd449ca77a.jpg: 640x640 10 heads, 8.3ms
image 158/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2327_png.rf.91166b6940fc2d96ebd252159e0e896f.jpg: 640x640 3 helmets, 7.7ms
image 159/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2369_png.rf.67609812a4bce16c48fca34b381d6426.jpg: 640x640 1 helmet, 9.8ms
image 160/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2384_png.rf.2363641d1ca3858f7498c5870ed00351.jpg: 640x640 1 helmet, 7.6ms
image 161/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2395_png.rf.617111066dd7733925332795ce9ae7c7.jpg: 640x640 6 helmets, 7.7ms
image 162/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers23_png.rf.a6b30faed83a1650ae4e05b4f9beedb5.jpg: 640x640 3 helmets, 7.9ms
image 163/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2402_png.rf.8b2c581dab6e8e0eef25e774396b7b30.jpg: 640x640 4 helmets, 8.8ms
image 164/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2415_png.rf.2bee50412ef455c45c70be4d5b10bdad.jpg: 640x640 8 heads, 8 helmets, 8.7ms
image 165/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2428_png.rf.4cbe004de00d183cbaea84c60257622e.jpg: 640x640 8 heads, 8.7ms
image 166/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2439_png.rf.f62b00b352d571322dace965e87f9c81.jpg: 640x640 2 helmets, 13.2ms
image 167/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2465_png.rf.15bd3c960c0fffd4ae9bfe864c7d8374.jpg: 640x640 9 helmets, 7.8ms
image 168/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2467_png.rf.58a314c07620310bb474a09fdbbfe1fe.jpg: 640x640 4 helmets, 8.4ms
image 169/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2485_png.rf.92aff1fc2f48c1ee66a3c3f6124f2380.jpg: 640x640 2 helmets, 7.9ms
image 170/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2493_png.rf.8eabe771f14348b292b32f5daefc1a31.jpg: 640x640 8 helmets, 7.5ms
image 171/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers249_png.rf.14a7a56f31f9b2bb4ee71a994bd29c8a.jpg: 640x640 4 helmets, 8.6ms
image 172/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2503_png.rf.f71aa2640c23a7f7969fd354cd0a5c5f.jpg: 640x640 12 heads, 10.1ms
image 173/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2504_png.rf.df9aca80b91468a8ac441723dfb232d1.jpg: 640x640 9 heads, 2 helmets, 7.9ms
image 174/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2509_png.rf.5f25ce703d122770dcc7b01c7f02ffed.jpg: 640x640 9 helmets, 7.5ms
image 175/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers251_png.rf.3831d68bc32a9d882305e6982bacbe32.jpg: 640x640 3 helmets, 7.4ms
image 176/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2529_png.rf.b199d8a0ad180c78c69112691bd7da64.jpg: 640x640 6 heads, 7.4ms
image 177/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2540_png.rf.14a9a5a771bb09568039512b197d1365.jpg: 640x640 5 helmets, 8.0ms
image 178/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2556_png.rf.3c3b15488a4ef73ffd3dd3e79f0df67b.jpg: 640x640 2 helmets, 7.5ms
image 179/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2578_png.rf.65d89ed1bf23e7fa75cf6dd6a0c7fe34.jpg: 640x640 1 helmet, 8.0ms
image 180/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2585_png.rf.e45aaec2d3c247797d47b07041c69237.jpg: 640x640 3 helmets, 7.5ms

image 181/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2595_png.rf.cfa3ba593deae7f39981c2c09156681.jpg: 640x640 2 helmets, 9.1ms
image 182/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2596_png.rf.9e8593ee10e1402be4bfff485e4a85ec2.jpg: 640x640 7 helmets, 14.8ms
image 183/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2601_png.rf.070a5299cb81a604c65ddbb906d88c00.jpg: 640x640 4 helmets, 7.6ms
image 184/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2616_png.rf.2489e50026d2aeaebbf6c6239cc63001d.jpg: 640x640 7 heads, 7.5ms
image 185/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2622_png.rf.46c13fb30e6401f9b6fa77f887dab64e.jpg: 640x640 4 helmets, 8.4ms
image 186/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2624_png.rf.4757c890d98ca71fa672cf953229ddad.jpg: 640x640 4 helmets, 7.8ms
image 187/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2640_png.rf.97664dabeb8487c664db23e161a3ee08.jpg: 640x640 14 helmets, 7.4ms
image 188/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2644_png.rf.595121a5a5cc91c5e8670858f979214e.jpg: 640x640 7 helmets, 7.4ms
image 189/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2647_png.rf.afff3c5e7bea091fe807df702a686bb2.jpg: 640x640 4 helmets, 8.1ms
image 190/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2656_png.rf.aaefd2e80f653b473d9b6b90399982b3.jpg: 640x640 9 helmets, 7.6ms
image 191/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2672_png.rf.34422ddd573b49f1d644f0a3d9d39179.jpg: 640x640 3 helmets, 7.3ms
image 192/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2692_png.rf.5f2d77c2877b718d6de571497d94bf0e.jpg: 640x640 8 heads, 7.3ms
image 193/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2695_png.rf.dfcd9d3ce3e8a156285b8a38e1d4a1c6.jpg: 640x640 4 helmets, 9.1ms
image 194/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2698_png.rf.334ec700a8d7f65720d3dd83805d0dd0.jpg: 640x640 2 helmets, 7.2ms
image 195/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2709_png.rf.b93cfd538b727bdb888d29a958492d26.jpg: 640x640 2 helmets, 18.9ms
image 196/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2711_png.rf.d4f937d543b37e0dcb0032ba81f9097f.jpg: 640x640 3 helmets, 18.4ms
image 197/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2723_png.rf.f3f501007fe17f6a519c76612a87e4b0.jpg: 640x640 3 helmets, 16.2ms
image 198/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2726_png.rf.399bbe57df2b8f0d130fa9bb8643d0a6.jpg: 640x640 3 helmets, 18.7ms
image 199/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers272_png.rf.a178fba41be52b430bf1110b7cab3622.jpg: 640x640 1 helmet, 11.7ms
image 200/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2734_png.rf.cc5b25b8e3481964695a194cb6cf8117.jpg: 640x640 2 heads, 3 helmets, 16.9ms
image 201/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers273_png.rf.a3e5d2c5a21710026742c41e5c9ebf6f.jpg: 640x640 2 helmets, 15.2ms
image 202/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2750_png.rf.510e8e686262446034a9bbfd09033755.jpg: 640x640 3 helmets, 36.8ms
image 203/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2752_png.rf.ac579d0443ccb2f6cae6b4d24c8f26cd.jpg: 640x640 4 helmets, 15.5ms
image 204/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2758_png.rf.2970c25374f9b99c93164f954ee68b0a.jpg: 640x640 2 helmets, 18.3ms
image 205/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2762_png.rf.2f1a91cec223ae150ff86fac0d16b00e.jpg: 640x640 5 helmets, 16.0ms
image 206/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2772_png.rf.f3d2d3ec4523aa08f7a9ae81452bfd9a.jpg: 640x640 3 helmets, 10.5ms
image 207/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2812_png.rf.823b4aabce9c5e3e6d9f048ec39e9b67.jpg: 640x640 1 head, 11 helmets, 12.6ms
image 208/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2815_png.rf.0944b49b3f2f57a740dbf3d08065b4d9.jpg: 640x640 1 head, 6 helmets, 11.8ms
image 209/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2824_png.rf.11fbd88118cfbfc0be510bd879c1f392.jpg: 640x640 1 helmet, 10.4ms
image 210/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2826_png.rf.40722e1fc3c3dbd7f54868ab979af95d.jpg: 640x640 2 helmets, 11.6ms

image 211/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2837_png.rf.8ed65a84dd85265824528bf30cc33883.jpg: 640x640 3 helmets, 10.1ms
image 212/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2882_png.rf.b548eb33a9353851132263eef9948514.jpg: 640x640 1 helmet, 9.9ms
image 213/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2884_png.rf.bd87318ee6685b8a603786f9c86956b4.jpg: 640x640 6 helmets, 10.0ms
image 214/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2898_png.rf.04f33981e24bcfceedd0c019195c5a22.jpg: 640x640 4 helmets, 14.4ms
image 215/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2924_png.rf.e96978594d4e2d78ac5e9428a092500e.jpg: 640x640 3 helmets, 11.4ms
image 216/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2935_png.rf.3d2e80a13ff34f9eb9cb38618851d868.jpg: 640x640 9 heads, 9.7ms
image 217/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2936_png.rf.f9693d691d5040e668a74603ed6a7b0c.jpg: 640x640 1 head, 9 helmets, 10.0ms
image 218/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2939_png.rf.6cb140982e68807a9e9e727471f9cce1.jpg: 640x640 15 helmets, 9.9ms
image 219/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers295_png.rf.4a02221ad58cd57dcd790315fc0b7a33.jpg: 640x640 16 heads, 6 helmets, 14.5ms
image 220/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2965_png.rf.25a1bfff3776adfd843bb9f0deeab72db.jpg: 640x640 3 helmets, 13.2ms
image 221/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2977_png.rf.c13102b806250b3d3292b01e742d0473.jpg: 640x640 5 heads, 12.8ms
image 222/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers297_png.rf.c4f1afb46c5a52b6c2c4aad0da006b12.jpg: 640x640 5 helmets, 11.5ms
image 223/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2982_png.rf.bdc038121d4d2d0203028ab884432970.jpg: 640x640 4 helmets, 10.2ms
image 224/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2986_png.rf.211a9229f2cb559b17928da5e02b7f58.jpg: 640x640 4 helmets, 11.8ms
image 225/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2990_png.rf.9bdb95f6c90270c74b697cb6f9ed71a7.jpg: 640x640 10 heads, 2 helmets, 10.5ms
image 226/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2992_png.rf.23ab5aa1c0444d22bdcc74c66104710e.jpg: 640x640 1 helmet, 10.0ms
image 227/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2994_png.rf.ddb38db02960871656686f1c1275041e.jpg: 640x640 11 heads, 9.5ms
image 228/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers2997_png.rf.f4a9bc2741b93e75aec6cc499aa11c98.jpg: 640x640 2 helmets, 9.8ms
image 229/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3000_png.rf.3a968d2d2103704b1723f69ad0bbb416.jpg: 640x640 7 helmets, 9.9ms
image 230/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3013_png.rf.4279b5867565776a022d3e4e7c122d6b.jpg: 640x640 4 heads, 4 helmets, 10.3ms
image 231/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3018_png.rf.e525de1996364997021d2b911a7b0c82.jpg: 640x640 3 helmets, 10.4ms
image 232/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers302_png.rf.6769f45cdd32c678cf5fbb64f3283039.jpg: 640x640 6 heads, 12.2ms
image 233/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3083_png.rf.104c078d77cbf57f13123bc00c5ffdc7.jpg: 640x640 2 helmets, 10.5ms
image 234/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers30_png.rf.143788c21b229a2f0fa4a0dcc128fb99.jpg: 640x640 3 helmets, 13.0ms
image 235/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3111_png.rf.12d3021ece4de10ef50f1e395f6ade12.jpg: 640x640 2 helmets, 28.4ms
image 236/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3114_png.rf.d08d930b8e2e41744eee0236ddb3d1f7.jpg: 640x640 1 head, 3 helmets, 27.2ms
image 237/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3120_png.rf.0afafb981afc7ebe16f4e162aaf1c8c4.jpg: 640x640 1 helmet, 36.5ms
image 238/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3129_png.rf.a646b914eedf0d51aead14300a82867f.jpg: 640x640 1 helmet, 13.9ms
image 239/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3136_png.rf.b28f11efc81e40103b828dc8dcff4f3b.jpg: 640x640 2 helmets, 11.8ms
image 240/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3176_png.rf.5280cd214cdfb5e3ab4f9ed35ed0eefa.jpg: 640x640 3 helmets, 38.3ms

image 241/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3187_png.rf.b63dae9f7e9d9a40b25a38f240110128.jpg: 640x640 2 helmets, 12.7ms
image 242/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3191_png.rf.892000500d6ad4f90f336ec7514fb493.jpg: 640x640 6 helmets, 11.1ms
image 243/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3195_png.rf.3e64e539e53d95f772f5fe00af4dd89e.jpg: 640x640 3 helmets, 9.6ms
image 244/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3205_png.rf.a2abc2580257c71807f5f7d2e40a3803.jpg: 640x640 7 helmets, 9.7ms
image 245/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3212_png.rf.87d0e507ee3cb069f081db68b54f9aac.jpg: 640x640 1 head, 3 helmets, 9.2ms
image 246/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3232_png.rf.36247e4f65e2c079d5562a0a06cd2810.jpg: 640x640 2 helmets, 9.4ms
image 247/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers324_png.rf.17f791c5507863ce754d2026d9857c45.jpg: 640x640 6 heads, 1 helmet, 10.1ms
image 248/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3272_png.rf.3fe07d81f8e5ff6849c3d9dadbe7ed19.jpg: 640x640 4 helmets, 10.3ms
image 249/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3281_png.rf.b1a30404e058dd575b4bf1088d5bc788.jpg: 640x640 6 helmets, 9.7ms
image 250/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers328_png.rf.59115d2cddad31bd963d8440924741de.jpg: 640x640 6 helmets, 9.4ms
image 251/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3292_png.rf.c141bca29850ad88431b5de6b178aa5f.jpg: 640x640 3 helmets, 22.3ms
image 252/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3318_png.rf.2dad432c80cd35c9836cc897bb0e6ae.jpg: 640x640 8 helmets, 26.4ms
image 253/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3320_png.rf.e5bfba7d089e169b189c1fa1128b2cb3.jpg: 640x640 2 helmets, 20.8ms
image 254/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3323_png.rf.dfed85b270b933e6269d97483d526025.jpg: 640x640 4 helmets, 27.1ms
image 255/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3329_png.rf.2b7f56da0423a7d22d179b9517928db1.jpg: 640x640 15 heads, 2 helmets, 30.4ms
image 256/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3332_png.rf.624a6b9afea70f79fd33529b8914e1de.jpg: 640x640 2 helmets, 56.2ms
image 257/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3347_png.rf.fabeb8c40c743a7006e07ef28abfe582.jpg: 640x640 8 helmets, 23.6ms
image 258/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers334_png.rf.83076d513ddf8fed8053e126d7ef9289.jpg: 640x640 2 helmets, 30.7ms
image 259/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3371_png.rf.b61fd8de33b2742cda8fdc8a4bd52521.jpg: 640x640 3 helmets, 32.8ms
image 260/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3378_png.rf.eedc1f2739bd3c37d8b1ccb31e09060b.jpg: 640x640 1 head, 3 helmets, 29.5ms
image 261/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3388_png.rf.abe6e37764762b7c793497b39b4119d0.jpg: 640x640 3 helmets, 31.9ms
image 262/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3390_png.rf.ba81498ac135298ef02ef3c59c51b2cc.jpg: 640x640 4 helmets, 19.0ms
image 263/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3392_png.rf.8a295eb06ae05fafa92b6a02cc767deb.jpg: 640x640 4 helmets, 14.2ms
image 264/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3405_png.rf.3a4d0ba9c7498a4cabf61868ff001c2b.jpg: 640x640 1 helmet, 28.4ms
image 265/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3409_png.rf.ffc4ebdcaab2242a0f9dd4d898d367a9.jpg: 640x640 4 helmets, 30.1ms
image 266/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3429_png.rf.25329f709062f35db82d8a15006e9218.jpg: 640x640 6 helmets, 19.0ms
image 267/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3438_png.rf.64f49d3859094268b7b7a53df7589716.jpg: 640x640 6 heads, 33.9ms
image 268/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3442_png.rf.b28e92eee3d2d3fb35d73c6dd1d6875b.jpg: 640x640 4 helmets, 17.5ms
image 269/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3446_png.rf.9672c73f29f10ea4c0b1419f4ef94e68.jpg: 640x640 2 helmets, 30.7ms
image 270/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers344_png.rf.72b8f954ac0468f6a9a7bc1958355a7e.jpg: 640x640 2 helmets, 35.5ms

image 271/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3459_png.rf.d0b4d420f5801295bab5df52e269dde3.jpg: 640x640 4 helmets, 57.2ms
image 272/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers345_png.rf.caab484e353080ee3bb9cbf0c863b9d8.jpg: 640x640 1 helmet, 32.9ms
image 273/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3465_png.rf.d7b1b95204f69ba03592fd1e0669da3f.jpg: 640x640 3 heads, 26.5ms
image 274/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers346_png.rf.55e6ec2a2d01f8b9e8fc0bd4d0c89808.jpg: 640x640 1 head, 15 helmets, 26.3ms
image 275/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3470_png.rf.85b55feb08d32e5fb039251a4d398af0.jpg: 640x640 2 helmets, 37.6ms
image 276/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3476_png.rf.850d369049ac954610abaad6a02beb0a.jpg: 640x640 1 helmet, 27.6ms
image 277/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3509_png.rf.11ef37271e4e4f095f3ebf560515b5c0.jpg: 640x640 2 helmets, 34.6ms
image 278/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers350_png.rf.159e81b8dba649cb13869caed336cd6a.jpg: 640x640 5 helmets, 49.5ms
image 279/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3519_png.rf.ff0cc08251dac5d50d01bce4d17275b1.jpg: 640x640 5 heads, 2 helmets, 15.8ms
image 280/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3542_png.rf.e7c8ffc493a1360a741cb6ce8e407909.jpg: 640x640 1 helmet, 30.2ms
image 281/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3558_png.rf.18d48a76355a2e5f10e5442a656b68ce.jpg: 640x640 2 helmets, 14.5ms
image 282/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3560_png.rf.e6f684076785b402073dcb9c1fd576bf.jpg: 640x640 4 helmets, 16.7ms
image 283/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3561_png.rf.01be126fbb827a23bbc8107c307b91af.jpg: 640x640 3 helmets, 15.3ms
image 284/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3598_png.rf.70be5b9239a1ad51160d2b8d9f63a862.jpg: 640x640 14 helmets, 11.7ms
image 285/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers359_png.rf.4289850011ca9a5b0fb2e3da9fd23b2b.jpg: 640x640 2 helmets, 7.9ms
image 286/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3607_png.rf.8f9ba58ed9cec2396a38eebc370077cf.jpg: 640x640 1 helmet, 7.8ms
image 287/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3612_png.rf.582fe4db506486d24c0e60fbec7c1050.jpg: 640x640 1 helmet, 7.8ms
image 288/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3615_png.rf.e644276627e4c396346fae2a411bc616.jpg: 640x640 9 helmets, 7.8ms
image 289/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers361_png.rf.aee9f418ae9def3b269a61565a200fbd.jpg: 640x640 3 helmets, 9.9ms
image 290/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers362_png.rf.21cf3d36b228d04c9ef8dcc51e1109ca.jpg: 640x640 2 helmets, 7.8ms
image 291/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3641_png.rf.3436ee4e3c409747c68da74d8979ab7c.jpg: 640x640 3 helmets, 7.8ms
image 292/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3642_png.rf.f3903ce1dfd6e7d60a9a6c534b059fd7.jpg: 640x640 3 helmets, 7.7ms
image 293/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3648_png.rf.2d811dec9f805dea255505936eec9b6a.jpg: 640x640 5 helmets, 8.6ms
image 294/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3653_png.rf.9c0f5388c4fb3808e1af9d51edab1558.jpg: 640x640 1 helmet, 7.9ms
image 295/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3667_png.rf.edca7f80fa3282278d727eccf7119486.jpg: 640x640 1 helmet, 7.7ms
image 296/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3677_png.rf.dc7aa36071bc5d0782253505f2b29da2.jpg: 640x640 1 helmet, 7.7ms
image 297/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3680_png.rf.435a2ecd61f20d5ce1fa20d015241a00.jpg: 640x640 8 helmets, 7.7ms
image 298/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3684_png.rf.0248f738c51d7b230199ef35fc52a421.jpg: 640x640 18 heads, 8.6ms
image 299/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3696_png.rf.4b0b2af578d6e28f35e75e1b6c7f17ee.jpg: 640x640 2 helmets, 10.7ms
image 300/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3708_png.rf.b101ee4bc4bedac1a292750f2f8e4794.jpg: 640x640 2 helmets, 7.7ms

image 301/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3718_png.
rf.8c00f8b63f663280ae8ba9b81179fa0b.jpg: 640x640 2 helmets, 7.7ms
image 302/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3734_png.
rf.7b7df854659cc94bee88cf2e7c733976.jpg: 640x640 1 helmet, 7.7ms
image 303/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3737_png.
rf.3f675247d5028ccf74189089ea5d1923.jpg: 640x640 1 head, 6 helmets, 8.2ms
image 304/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3746_png.
rf.7eaab3a0f68836ea1e441281b8c88c73.jpg: 640x640 3 helmets, 10.1ms
image 305/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3751_png.
rf.4d17f3ce0bdc1b7651de53477504acf2.jpg: 640x640 1 head, 5 helmets, 7.2ms
image 306/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3752_png.
rf.df8029eeab6e265948c1a1068ab07d95.jpg: 640x640 3 helmets, 7.4ms
image 307/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3758_png.
rf.8c997ae8e2b104c9ed50aeba7c6b6442.jpg: 640x640 3 heads, 2 helmets, 7.5ms
image 308/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers375_png.r
f.c0a9bd0fc46e81cddcf4688c161f393b.jpg: 640x640 4 helmets, 13.3ms
image 309/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3762_png.
rf.7777e6a3b3427279e95605931c42bdf6.jpg: 640x640 13 helmets, 13.2ms
image 310/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3781_png.
rf.8744350a0c94db36df388da6dcda6973.jpg: 640x640 4 helmets, 8.1ms
image 311/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3788_png.
rf.7b5f156c05611db3296401b9c690b156.jpg: 640x640 2 helmets, 8.2ms
image 312/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3793_png.
rf.3ee7f0ec17c59d5d73117f3b6f55cdcd.jpg: 640x640 2 helmets, 8.4ms
image 313/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3807_png.
rf.e455458f45c785440466e90827a7c79c.jpg: 640x640 4 heads, 3 helmets, 7.8ms
image 314/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3811_png.
rf.ebc1637ef6fdf4b0857bc7f30b637f16.jpg: 640x640 2 helmets, 9.8ms
image 315/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3840_png.
rf.26a3950853c0aa00a42454d234f53ce6.jpg: 640x640 6 heads, 2 helmets, 7.9ms
image 316/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3866_png.
rf.a2ef2d3d1414f739997cac61bbb11949.jpg: 640x640 1 helmet, 13.2ms
image 317/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3874_png.
rf.1321ba3ed6e30ec9dee95b0e555f585f.jpg: 640x640 2 helmets, 8.0ms
image 318/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3881_png.
rf.7d9f6bf43562986149cf14cef2faa08.jpg: 640x640 10 helmets, 8.3ms
image 319/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3887_png.
rf.fdf79f3e35748bdaada1471b841e06c8.jpg: 640x640 16 helmets, 7.8ms
image 320/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3889_png.
rf.55da258957fd56bcfde779dfe53bf905.jpg: 640x640 4 helmets, 7.8ms
image 321/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3891_png.
rf.41e4bac6efd5375dc76464bcd9b5cd9b.jpg: 640x640 1 helmet, 7.2ms
image 322/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3893_png.
rf.ce639dc9f1bf9e7a81dbc23eb0cd98d6.jpg: 640x640 1 helmet, 7.7ms
image 323/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3905_png.
rf.b55c7d9b01fa986e94e8edeec91ea344.jpg: 640x640 3 helmets, 7.5ms
image 324/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3907_png.
rf.ceca41d3940525c03d222f664ccd6fb5.jpg: 640x640 11 helmets, 8.1ms
image 325/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3913_png.
rf.027817adcaaeccc9f0b68e06d11ca3e7.jpg: 640x640 4 helmets, 9.7ms
image 326/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3930_png.
rf.cc79d5c7fb5e13a96d10c1689ca31af6.jpg: 640x640 2 helmets, 7.4ms
image 327/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3937_png.
rf.1ac154892e32b25f2ff5e50aa214a072.jpg: 640x640 4 helmets, 7.4ms
image 328/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3942_png.
rf.995f18ac2d75476731922d7dc2892e49.jpg: 640x640 3 helmets, 7.4ms
image 329/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3945_png.
rf.8fc98430df5c9e16b4e686408bf312c7.jpg: 640x640 12 heads, 8.6ms
image 330/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3946_png.
rf.b8087021bad08b52b3f5796822bdb26c.jpg: 640x640 1 head, 10 helmets, 8.5ms

image 331/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3953_png.rf.f6eaf9d9ce455b185497b10e3d98c311.jpg: 640x640 3 heads, 1 helmet, 8.4ms
image 332/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3966_png.rf.5bca29b5a0a7e8a12a73ec0d475735df.jpg: 640x640 5 helmets, 8.6ms
image 333/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3977_png.rf.725434a7faa07748a1143bda7ea61bb7.jpg: 640x640 5 helmets, 7.8ms
image 334/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3986_png.rf.94b52b12b51c243eb2a67a16918bea11.jpg: 640x640 4 helmets, 7.8ms
image 335/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers398_png.rf.18a484426e00d03b1218fd4ddcced608.jpg: 640x640 4 helmets, 7.6ms
image 336/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers3994_png.rf.d0768422871fa7601c3ba990e832fcf6.jpg: 640x640 16 helmets, 8.7ms
image 337/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4000_png.rf.0a01718008c02ffdfa0eaa5e4a1f7c54.jpg: 640x640 4 helmets, 7.9ms
image 338/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4004_png.rf.ed4f51ddb3647fa45baeda2049f4e167.jpg: 640x640 6 helmets, 7.4ms
image 339/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4008_png.rf.4648481aeb0b74997170de5fa59624d6.jpg: 640x640 6 helmets, 12.9ms
image 340/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4009_png.rf.4c47e333f50e065bb6b0dd2661c03e77.jpg: 640x640 6 helmets, 14.5ms
image 341/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4029_png.rf.f6779e8431af003f47be7cb1cce0f8ee.jpg: 640x640 2 helmets, 13.5ms
image 342/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4031_png.rf.23cf04a87846d53e926e7406fc6f4ed4.jpg: 640x640 3 helmets, 17.1ms
image 343/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4036_png.rf.926f640f624426f61e7df6d417fb250c.jpg: 640x640 5 heads, 14.9ms
image 344/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4045_png.rf.32951a239a57fae38b38c0b41a2e9486.jpg: 640x640 3 helmets, 16.9ms
image 345/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4050_png.rf.e3849eaaa12311f891d7df617e1d5b6f.jpg: 640x640 9 helmets, 13.9ms
image 346/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4054_png.rf.e0a343fa780b0a73c33a3faa83ad8b03.jpg: 640x640 2 helmets, 17.0ms
image 347/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4057_png.rf.79e2382f071d26807ca4e944b7f5e336.jpg: 640x640 4 helmets, 25.3ms
image 348/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4058_png.rf.c387e46e64b91d422f471f2e5cee01e7.jpg: 640x640 5 helmets, 24.1ms
image 349/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers405_png.rf.6f81da244797e0040d4259b20b093f9c.jpg: 640x640 3 helmets, 19.1ms
image 350/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4067_png.rf.996f0c027c1b72f39b5f8416123f00da.jpg: 640x640 2 heads, 2 helmets, 22.4ms
image 351/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4070_png.rf.17f5795cd85ac03ef982dc50aa0e2ed0.jpg: 640x640 4 helmets, 18.7ms
image 352/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4073_png.rf.a913e05c484721cd5cb00f7ce7df9883.jpg: 640x640 7 helmets, 76.6ms
image 353/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4076_png.rf.255f62c1fc5a4e8094d88542bc19e409.jpg: 640x640 12 heads, 19.7ms
image 354/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4080_png.rf.8b894d96854948bc41dc804a82335b35.jpg: 640x640 3 helmets, 22.1ms
image 355/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4090_png.rf.441f1c48831a6dc7e9c8e9c176793977.jpg: 640x640 5 helmets, 37.3ms
image 356/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4095_png.rf.0220dea18cb85ee58d6edfe435a10e5f.jpg: 640x640 2 helmets, 15.8ms
image 357/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4097_png.rf.8db5476d7a524a44d14fe21b58ba37c3.jpg: 640x640 4 helmets, 16.3ms
image 358/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4100_png.rf.cad1f6d04535bc64e9511c2693a6229f.jpg: 640x640 4 helmets, 15.1ms
image 359/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4105_png.rf.78c23880ef08e4b7f3d4928ee28859e3.jpg: 640x640 3 helmets, 14.8ms
image 360/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4107_png.rf.89e4bbbedeea8e549e93008f73397ec0.jpg: 640x640 11 helmets, 33.0ms

image 361/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4119_png.rf.1f5f1ca9a6e3737b205f68ca1cb504d4.jpg: 640x640 3 helmets, 16.3ms
image 362/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4134_png.rf.7e686edb9fd429af7d0435c43546ad86.jpg: 640x640 6 helmets, 13.7ms
image 363/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4136_png.rf.092c614ff36d0e477a16d9ae2946f26a.jpg: 640x640 12 heads, 6 helmets, 25.1ms
image 364/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4139_png.rf.66d4aa71e0013a733a97f8f72fc2df74.jpg: 640x640 3 helmets, 36.1ms
image 365/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4171_png.rf.63489d749e0e566d495af0f1c7af20ee.jpg: 640x640 6 helmets, 46.2ms
image 366/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers41_png.rf.11617019b8e61d5c8d5b4e1a4e9d0253.jpg: 640x640 2 helmets, 43.6ms
image 367/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4201_png.rf.d78f5befe3fab36e50b5ee53501b80af.jpg: 640x640 6 helmets, 31.7ms
image 368/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4214_png.rf.86a1db895c70890c5873ddd9cb774703.jpg: 640x640 2 helmets, 21.7ms
image 369/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4217_png.rf.01163aca3699632b2209c7fc88ced0dc.jpg: 640x640 7 helmets, 32.3ms
image 370/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4219_png.rf.7d8bb98b61326a2dcb88f0f9fe68e185.jpg: 640x640 2 helmets, 11.1ms
image 371/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4231_png.rf.ddb29174048f450777dbd965b823e46f.jpg: 640x640 6 heads, 7.8ms
image 372/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4265_png.rf.003e7eef4d0d9f062e3dfdcacc4fdcf0.jpg: 640x640 6 helmets, 7.8ms
image 373/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4274_png.rf.15a8de6c2d4d51be59f12e3f080084dd.jpg: 640x640 3 helmets, 7.8ms
image 374/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers428_png.rf.d7ca1cc8a008a482267192ef0a2bda02.jpg: 640x640 3 helmets, 7.8ms
image 375/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4292_png.rf.8090846cd430e563416a02ecf882693e.jpg: 640x640 4 helmets, 9.5ms
image 376/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4293_png.rf.7b217bf7853c7ccd5963e2b9a2a0d0b4.jpg: 640x640 2 helmets, 9.6ms
image 377/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4296_png.rf.4044ccd7373fe735be94c2b74e0ca41c.jpg: 640x640 1 head, 3 helmets, 7.8ms
image 378/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4299_png.rf.3aa0ac87a57246043e1fc91d78d689c9.jpg: 640x640 3 helmets, 7.8ms
image 379/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4314_png.rf.7188660157ab08550ee601786029a678.jpg: 640x640 2 helmets, 7.8ms
image 380/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4315_png.rf.e5d0d9ab70f70f4a3c8d4a301de07269.jpg: 640x640 4 helmets, 7.4ms
image 381/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4320_png.rf.38e14a738a77a4d007997058d2b571dd.jpg: 640x640 8 heads, 7.4ms
image 382/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4321_png.rf.af79c9376697b1ac0a550835ecc65360.jpg: 640x640 2 helmets, 10.7ms
image 383/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4324_png.rf.ebe8e7fadcf2f1b6b9c51c0f37dde0085.jpg: 640x640 5 helmets, 7.4ms
image 384/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers432_png.rf.9e99fcf630e3c120d3304947faf86c9b.jpg: 640x640 2 helmets, 7.6ms
image 385/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4361_png.rf.7d9f16e15d7be729b9a5192beec6913.jpg: 640x640 12 heads, 7.4ms
image 386/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4371_png.rf.49414e9271e028ac3af50e900137ffa5.jpg: 640x640 5 heads, 7.4ms
image 387/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4382_png.rf.5e976af4c5e921883d60ebee4751939d.jpg: 640x640 4 helmets, 7.9ms
image 388/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4383_png.rf.6be8b539216436f3aa949e1b5e9963e7.jpg: 640x640 24 heads, 9.4ms
image 389/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers438_png.rf.0c25e935198badbd05ead9b38de81be.jpg: 640x640 2 helmets, 7.4ms
image 390/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4391_png.rf.7130cdf7725ffd02ca925645f33df56c.jpg: 640x640 5 helmets, 7.4ms

image 391/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4398_png.rf.5757571253446cf5bc522f2f9bd79f93.jpg: 640x640 5 helmets, 7.0ms
image 392/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4400_png.rf.c3d31302b3a9d27bd676b38d550f6bbe.jpg: 640x640 2 helmets, 8.4ms
image 393/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4428_png.rf.83770d0e588f6eb9fcfd96e8e27321.jpg: 640x640 4 helmets, 7.1ms
image 394/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4454_png.rf.81661f67f59a50b4993aa3a0562ea19c.jpg: 640x640 11 heads, 7.5ms
image 395/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers445_png.rf.d460aaa74b4ee59d8bffb3937365b1dd.jpg: 640x640 2 helmets, 7.4ms
image 396/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4465_png.rf.9ab50673b275cb77d8dd83eb84b8ee6b.jpg: 640x640 2 helmets, 9.8ms
image 397/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4491_png.rf.528c12dbf2a3e85dd986b13c3fc2e47c.jpg: 640x640 2 helmets, 7.6ms
image 398/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4505_png.rf.bb382e994820084fc768ab47409825f.jpg: 640x640 2 helmets, 7.8ms
image 399/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers450_png.rf.90b743da517f8e8b3d20ae08845ed50a.jpg: 640x640 6 helmets, 9.4ms
image 400/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4515_png.rf.f5514213f79981fe1ed480fd18607c49.jpg: 640x640 1 helmet, 8.5ms
image 401/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4516_png.rf.529b509991e66602ccc152c6f2337496.jpg: 640x640 2 helmets, 7.3ms
image 402/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4526_png.rf.b2b852cd445a5ff0ef026a1f86aa3314.jpg: 640x640 2 helmets, 7.5ms
image 403/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4561_png.rf.f447f2974550fa57730c7cdee3eeaa33.jpg: 640x640 3 helmets, 8.0ms
image 404/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4570_png.rf.18257695499d5a06885c1f99acb00510.jpg: 640x640 8 heads, 7.3ms
image 405/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4572_png.rf.33caaffc8f3aca043a9707502a35d9d8.jpg: 640x640 1 head, 1 helmet, 8.8ms
image 406/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4577_png.rf.c4c3bf9e7075f8bc2ec2c62bc0aa92ff.jpg: 640x640 2 helmets, 6.9ms
image 407/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4587_png.rf.24a73c45a2117771d5a4cf038f61469c.jpg: 640x640 1 helmet, 7.0ms
image 408/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4588_png.rf.c9f2e3c94c676d213841d84ca495a6e0.jpg: 640x640 9 helmets, 7.1ms
image 409/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4596_png.rf.29c20eb4196988a661fa13face683e3e.jpg: 640x640 18 heads, 3 helmets, 7.0ms
image 410/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4609_png.rf.81e5f0bda6a0037164f59e6c5d658d1e.jpg: 640x640 1 helmet, 7.0ms
image 411/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers460_png.rf.5d715efb4fae4f13eeee6154c4339793.jpg: 640x640 7 heads, 7.9ms
image 412/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4614_png.rf.5654626109656d213ba907976185e628.jpg: 640x640 8 helmets, 7.3ms
image 413/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4617_png.rf.0b80cbe60764aded9272c942a5759bc5.jpg: 640x640 29 heads, 1 helmet, 7.7ms
image 414/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4623_png.rf.ef51446ec0f2213b7e7ae36d082a45d4.jpg: 640x640 6 heads, 2 helmets, 7.8ms
image 415/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4625_png.rf.c71b555600eaf9259338736cb69d0ca1.jpg: 640x640 9 heads, 7.4ms
image 416/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4627_png.rf.591c8fb6f98bb90eea1a7650c946e08f.jpg: 640x640 3 helmets, 7.0ms
image 417/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4649_png.rf.ade8c29a49a17ffa579ad3761270e098.jpg: 640x640 1 helmet, 8.4ms
image 418/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4656_png.rf.03032e93204e8f343adc0d6171f7782e.jpg: 640x640 9 heads, 6.9ms
image 419/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4668_png.rf.75dd2ad52a34f1a081c2c7c753ca5c62.jpg: 640x640 5 helmets, 7.1ms
image 420/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4687_png.rf.1bbd62b115becebeebc1b470d4ba86f7.jpg: 640x640 1 helmet, 7.1ms

image 421/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4695_png.rf.9970467bcc4ec64a2b3afa09eaefb3b9.jpg: 640x640 2 helmets, 7.1ms
image 422/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4701_png.rf.11612b574c0c8c40e96e50506e58d1be.jpg: 640x640 5 helmets, 7.0ms
image 423/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4717_png.rf.c9ac6c057ed5080225510ac81579cb26.jpg: 640x640 8 helmets, 7.1ms
image 424/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4720_png.rf.b76e19068dcd4a296103c511ce4b92eb.jpg: 640x640 1 helmet, 7.2ms
image 425/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4725_png.rf.50066ffbf0df66f4be42f29f37c3fa2c.jpg: 640x640 2 helmets, 8.6ms
image 426/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4732_png.rf.fa52b006d036c7197916065230010dd9.jpg: 640x640 3 heads, 16 helmets, 14.0ms
image 427/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4742_png.rf.1b5c6d9d47f48878d19f2de4d25d289c.jpg: 640x640 2 helmets, 7.3ms
image 428/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4747_png.rf.1e41e96741930a52377f75e8aff3dfca.jpg: 640x640 4 helmets, 11.6ms
image 429/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4771_png.rf.fcdb9a573c79ed31cc81d8adc40803c5.jpg: 640x640 2 helmets, 7.2ms
image 430/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4787_png.rf.cd2b45f18d70a6fd68e049be900a89dd.jpg: 640x640 8 heads, 7.3ms
image 431/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4789_png.rf.215670dc72be350cd1ffac9c118e40f1.jpg: 640x640 3 helmets, 7.6ms
image 432/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers478_png.rf.956968f27a2e032923656e5c7cbafd2b.jpg: 640x640 4 helmets, 7.2ms
image 433/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4792_png.rf.76555b8065fffc912499d1293f618b40d.jpg: 640x640 3 helmets, 7.4ms
image 434/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4795_png.rf.954060843f01a4520e7c5540f2886d7c.jpg: 640x640 30 helmets, 7.3ms
image 435/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4796_png.rf.6810363c6f478a46c7749e859fc57a35.jpg: 640x640 4 helmets, 7.1ms
image 436/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers47_png.rf.9c32b8abf999ed6a9c00f866ba298772.jpg: 640x640 2 helmets, 7.8ms
image 437/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4806_png.rf.129894a3568e7b20efe3fbaff606689a.jpg: 640x640 5 helmets, 7.2ms
image 438/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4808_png.rf.f7b2976413ee12d417e677e596aa391f.jpg: 640x640 5 helmets, 7.1ms
image 439/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4826_png.rf.e6b8cb3672150581a108dc5c50b53c9a.jpg: 640x640 8 helmets, 7.3ms
image 440/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4830_png.rf.a47aa995c66fab1455bab8fb0346d3c2.jpg: 640x640 1 helmet, 7.3ms
image 441/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4842_png.rf.c4d3c1680aa0dc0f4089c28a4dec73af.jpg: 640x640 9 heads, 2 helmets, 9.8ms
image 442/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers485_png.rf.9171c1bb4ae26c69eef9e07c57f3f43.jpg: 640x640 5 helmets, 8.4ms
image 443/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4866_png.rf.f8830b6b3c3c16f4b39abfc72c9c0add.jpg: 640x640 18 heads, 2 helmets, 7.2ms
image 444/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4872_png.rf.67dd3e13839ae83942313e3ad531cbdd.jpg: 640x640 2 helmets, 7.3ms
image 445/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4880_png.rf.aa6c2f5425aea14f42278d1d80f1de42.jpg: 640x640 11 helmets, 7.6ms
image 446/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4886_png.rf.70618745951330777a26c1b65b53f99d.jpg: 640x640 3 helmets, 9.7ms
image 447/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4895_png.rf.edab7df57cb032cc9d83bb33c87eac66.jpg: 640x640 15 helmets, 7.9ms
image 448/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4904_png.rf.3980471903f5f2f53761b160d5cdace6.jpg: 640x640 9 helmets, 7.6ms
image 449/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4909_png.rf.1bd097cab4403e867255e05fc508a6f8.jpg: 640x640 3 heads, 7 helmets, 7.6ms
image 450/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers490_png.rf.dea3cdab0c7df25b026cc913f83113aa.jpg: 640x640 31 heads, 5 helmets, 7.7ms

image 451/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4913_png.rf.93cfdff150c0171414efa03da189492a4.jpg: 640x640 2 heads, 8 helmets, 8.2ms
image 452/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4929_png.rf.83a849eb887eb6653529f7ed08a75395.jpg: 640x640 2 helmets, 8.2ms
image 453/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4950_png.rf.893321151017f44142b0e28b4b35a471.jpg: 640x640 2 helmets, 9.4ms
image 454/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4961_png.rf.4779dd240e759764fd885bdf3bcb8ac.jpg: 640x640 8 heads, 9.4ms
image 455/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4963_png.rf.54dcf9c0024fee020fbf77a8db9d62d9.jpg: 640x640 8 heads, 10.2ms
image 456/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4978_png.rf.9550c12f73f3ebbcda5c8eca41089b2d.jpg: 640x640 1 helmet, 7.9ms
image 457/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4985_png.rf.7022f3d989622557f6fbeb2ca2017554.jpg: 640x640 3 helmets, 7.7ms
image 458/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers4986_png.rf.4a22312e00e0dbcb1c01db312769916e.jpg: 640x640 1 helmet, 13.4ms
image 459/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers505_png.rf.6d1c490814d5a12e276dcfe8932f9992.jpg: 640x640 6 helmets, 7.8ms
image 460/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers511_png.rf.459dbd2d2d27858565b21fbe74b2a119.jpg: 640x640 2 helmets, 7.8ms
image 461/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers522_png.rf.66cdf28837b8cb57216054bae826304b.jpg: 640x640 7 heads, 7.7ms
image 462/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers528_png.rf.baa795d68a9cd2c7a9227b0f1e660a5c.jpg: 640x640 5 heads, 6 helmets, 7.9ms
image 463/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers529_png.rf.4cf46bd1992ece4be20d30c00e0ee215.jpg: 640x640 3 helmets, 8.8ms
image 464/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers536_png.rf.5c412d0873c896fa6f0fdbf19df00df6.jpg: 640x640 7 heads, 8.3ms
image 465/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers551_png.rf.3765a2b035d45aa8d64e16b9fc9213.jpg: 640x640 1 helmet, 7.2ms
image 466/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers564_png.rf.27dad6bfbfbbc8787526c433aac1128fc.jpg: 640x640 20 heads, 7.3ms
image 467/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers570_png.rf.9b6cf328971875eb921272a905849da2.jpg: 640x640 4 heads, 4 helmets, 7.2ms
image 468/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers574_png.rf.7d594e47c3771af765674a572c585404.jpg: 640x640 4 heads, 7.3ms
image 469/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers623_png.rf.8cb76afc532f76b6b39c892c43e4e688.jpg: 640x640 1 helmet, 9.8ms
image 470/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers630_png.rf.353af33626cb4485a17419b0a0cab615.jpg: 640x640 1 helmet, 7.2ms
image 471/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers632_png.rf.148c42f1905779020437c618a194de37.jpg: 640x640 5 helmets, 7.5ms
image 472/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers646_png.rf.7a766bfaf14b37cc5ef9f8837b5e2f5c.jpg: 640x640 4 helmets, 7.4ms
image 473/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers664_png.rf.b142f433a3ca63ca4cc564444d46e2e7.jpg: 640x640 1 helmet, 11.2ms
image 474/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers688_png.rf.f3aa060efe02f5de2cda5b0ccd9903e1.jpg: 640x640 2 helmets, 7.3ms
image 475/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers709_png.rf.1a3a2dda109256bde9dab3aecdd7242f.jpg: 640x640 2 helmets, 7.2ms
image 476/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers725_png.rf.962f084dd8bf8ac27ffe9c3e6866185b.jpg: 640x640 2 helmets, 7.4ms
image 477/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers739_png.rf.2da14d7985c6b9d1fd051a39ef12a5e7.jpg: 640x640 3 heads, 13 helmets, 9.2ms
image 478/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers749_png.rf.4a1304de956fe3bde4dcd9d09bb0941b.jpg: 640x640 2 helmets, 7.4ms
image 479/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers752_png.rf.9467ddbe8dafa1fd31cdfe667f0f0968.jpg: 640x640 37 helmets, 7.1ms
image 480/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers770_png.rf.9f51a1eb52f247d9c439b27262355964.jpg: 640x640 3 helmets, 10.3ms

```

image 481/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers773_png.r
f.c2701599d5afc27ffb480065cf862715.jpg: 640x640 1 helmet, 14.2ms
image 482/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers797_png.r
f.bf0995c0b22d0a2b0b1fd7eb173cce1f.jpg: 640x640 6 heads, 7.2ms
image 483/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers813_png.r
f.e9883011a77799305a38008841e74385.jpg: 640x640 2 helmets, 11.9ms
image 484/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers816_png.r
f.05af62e46e2414527b11abeb88f3a500.jpg: 640x640 3 helmets, 7.9ms
image 485/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers817_png.r
f.e286f20f1fb5c831062b8c4a92ca963c.jpg: 640x640 11 helmets, 7.1ms
image 486/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers832_png.r
f.b323df5f65a8f3adf296e5fdc083d0db.jpg: 640x640 8 helmets, 6.9ms
image 487/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers834_png.r
f.58eabc9175b8f0deed4a689fd7e305cc.jpg: 640x640 5 helmets, 7.1ms
image 488/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers840_png.r
f.cb22ccc1a4e4c0d232945e86543aa414.jpg: 640x640 4 helmets, 7.7ms
image 489/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers844_png.r
f.fd62776655b68498bc6ea777282b716b.jpg: 640x640 3 helmets, 7.4ms
image 490/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers846_png.r
f.5e593ac8855a0d94574eaa572f225e27.jpg: 640x640 2 heads, 11.1ms
image 491/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers868_png.r
f.adc40ec2d25a94421c56846e0767eb94.jpg: 640x640 1 head, 9 helmets, 8.0ms
image 492/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers870_png.r
f.8229707d8cc163493a7fc65615783d80.jpg: 640x640 2 helmets, 7.4ms
image 493/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers875_png.r
f.0f5ef25797623cb1cc7d8c3fd53aa1f9.jpg: 640x640 4 helmets, 7.3ms
image 494/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers88_png.r
f.4983d8b23341547a54673ad39e2cb1fa.jpg: 640x640 2 helmets, 7.6ms
image 495/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers902_png.r
f.3d9f040f92ce010f011568a9eb260bf5.jpg: 640x640 6 helmets, 7.3ms
image 496/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers931_png.r
f.23256a7be2ae33af235685978763bd2d.jpg: 640x640 1 head, 7 helmets, 9.1ms
image 497/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers961_png.r
f.87e48c2e85e6b2c5f5b0584072fb5c72.jpg: 640x640 8 helmets, 7.7ms
image 498/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers964_png.r
f.c4b748262a2104a425d4c7c297675448.jpg: 640x640 1 helmet, 7.2ms
image 499/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers965_png.r
f.ad41f5d475591cf2235423c3ab3ba5b4.jpg: 640x640 1 helmet, 8.4ms
image 500/500 /content/Hard-Hat-Detection-1/test/images/hard_hat_workers97_png.r
f.1cc5d0b489b42b2b4ec75af5c98db929.jpg: 640x640 7 helmets, 7.7ms
Speed: 2.7ms preprocess, 11.5ms inference, 1.6ms postprocess per image at shape
(1, 3, 640, 640)
Results saved to /content/runs/detect/predict

```

```

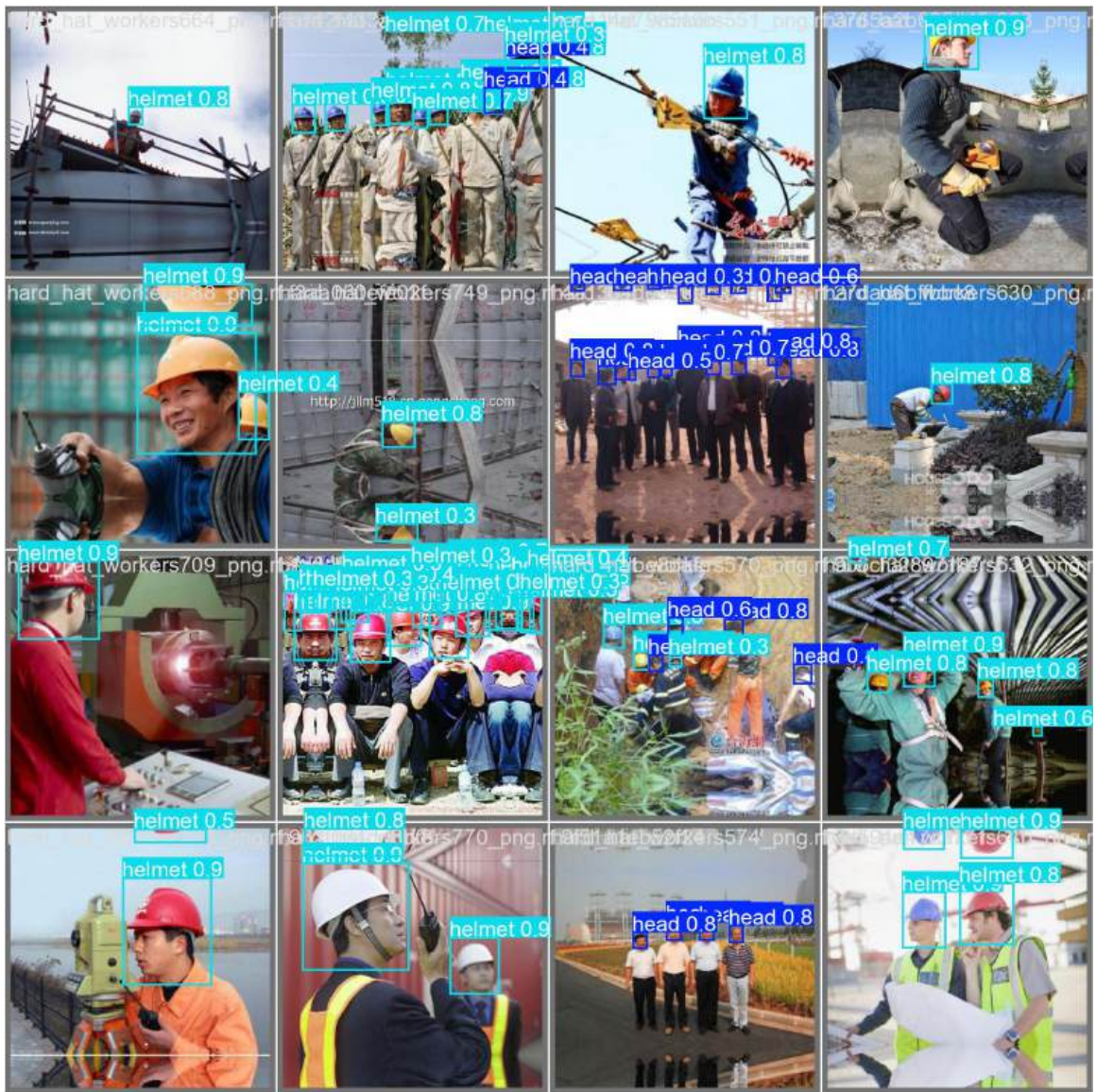
In [ ]: saved_image_path = os.path.join(results[0].save_dir, os.path.basename(results[0]
display(Image(saved_image_path))

```



```
In [ ]: Image('/content/evaluations/test_eval_v1/val_batch0_pred.jpg')
```


Out[]:



In []: Image('/content/evaluations/test_eval_v1/val_batch1_pred.jpg')

Out[]:



Dari hasil visual prediksi model:

a. Yang bagus:

- Model cukup konsisten mendeteksi helmet dan - head dengan confidence tinggi (0.7-0.9) di berbagai kondisi gambar
- Bounding box pada objek yang jelas terlihat cukup presisi
- Model mampu mendeteksi banyak objek sekaligus dalam satu frame

b. Yang bermasalah:

- Banyak bounding box yang overlap dan saling tumpuk saat objek berdekatan, sehingga label prediksi jadi susah dibaca
- Beberapa objek kecil atau yang jauh dari kamera confidence-nya rendah (0.3-0.4)
- Kelas person tidak terdeteksi sama sekali di semua gambar, konsisten dengan hasil evaluasi sebelumnya

Secara keseluruhan hasil visual sudah sesuai dengan metrics yang dianalisis. Model cukup baik untuk helmet dan head, namun gagal total untuk kelas person akibat data yang sangat sedikit dan tidak seimbang dibanding kelas lainnya.

Existing Challenges

- Class Imbalance. Distribusi kelas yang sangat tidak seimbang, terutama kelas person (~800) dibanding helmet (~19.000), menyebabkan model gagal mendeteksi kelas minoritas sama sekali.
- Bounding Box Kurang Presisi. Nilai mAP50-95 yang rendah (38.38%) menunjukkan model masih kesulitan menentukan posisi bounding box secara akurat, terutama pada objek kecil atau berdekatan.

Future Opportunities

- Menghapus atau Menyeimbangkan Kelas Person. menghapus kelas person atau menambah datanya secara signifikan supaya model bisa belajar lebih merata di semua kelas.
- Upgrade ke Model yang Lebih Besar. mencoba YOLOv11s atau YOLOv11m untuk meningkatkan kemampuan deteksi terutama pada objek kecil dan kondisi kompleks.
- Augmentasi Data. menambah variasi data training melalui augmentasi seperti perubahan pencahayaan, rotasi, dan skala untuk meningkatkan generalisasi model.
- Model ini bisa dikembangkan untuk sistem safety monitoring real-time di area konstruksi atau industri untuk memastikan pekerja menggunakan helmet dengan benar.

```
In [1]: !jupyter nbconvert --to html "/content/sample_data/Copy_of_Project_D1_FinalMaybe
```

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[NbConvertApp] Converting notebook /content/sample_data/Copy_of_Project_D1_FinalMaybe.ipynb to html
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[NbConvertApp] WARNING | Alternative text is missing on 16 image(s).
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[NbConvertApp] Writing 6142716 bytes to /content/sample_data/Copy_of_Project_D1_FinalMaybe.html
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In [ ]:
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